

Incentivizing Self-Protection From Wildfires

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Abstract

Losses from natural disasters can be mitigated through investments in property protection, yet adoption of low-cost risk-reducing measures remains low. In wildfire-exposed areas, managing vegetation and other fuels next to homes decreases structure vulnerability, but few properties comply with best practices despite widespread outreach efforts. We report on two randomized field experiments that incentivized thousands of Oregon homeowners to participate in pre-existing fire department programs to reduce wildfire risk. Across both experiments, financial incentives of \$100 to \$500 resulted in up to 12% of homeowners scheduling free defensible space assessments, while information-only approaches achieved assessment takeup rates of 3% at most. Responses to subsidies are largest among owners of more valuable homes, while baseline wildfire risk does not predict responsiveness. On-the-ground and remotely-sensed measures of vegetation provide noisy estimates of treatment effects on endline vegetation. Finally, we consider the extent to which the estimated behavioral elasticities from these experiments might shed light on potential homeowner responses to pricing of wildfire risk reductions in home insurance premiums, a practice that is becoming more common due to technological and regulatory changes.

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1 Introduction

As a share of GDP, natural disaster losses in the United States increased by a factor of five from the 1960s to the 2020s.¹ Efficient adaptation to these escalating risks will require many households to make cost-effective protective investments. For example, flood risk can be mitigated by sump pumps or flood shields, hurricane damage by storm shutters or roof tiedowns, and the health impacts of heat waves by air conditioning or cool roofs. Yet adoption of these technologies remains limited even in high-risk areas, reflecting a combination of unpriced positive externalities, risk misperception, moral hazard, and other frictions.²

This is especially true for wildfire risk, which is among the fastest-growing natural catastrophe perils.³ Research shows that managing vegetation and other flammable material near homes to establish “defensible space” can reduce wildfire damage at low cost (Cohen 2000; Syphard, Brennan, and Keeley 2014). Adoption of these practices remains limited despite widespread efforts by policymakers to promote defensible space.⁴ Compulsory approaches include vegetation codes, which are common on paper but rarely enforced, and recent efforts to promote differentiated pricing of homeowners insurance according to structure-level wildfire vulnerability (Ludington, Liao, and Walls 2025). As an alternative to these politically challenging policies, many local and state governments have instead relied on information campaigns and voluntary subsidy programs. Such programs play an increasing role in wildfire protection and climate change adaptation more broadly, but there are few, if any, formal evaluations of their design and effectiveness.

This paper reports on the findings of two randomized field experiments evaluating the effects of financial and non-financial interventions to promote defensible space. Both experiments were conducted in Oregon in close partnership with local fire districts and the Office of the Oregon State Fire Marshal. The first was in Jackson County Fire District 3 (JCFD3), a local fire protection district in Southern Oregon. The second was in the service territory of the Eugene-Springfield Fire department (ESF). Both experiments randomly assigned homes in at-risk areas to either a control group that received no communication, or to one of two treatment categories. The outreach-only treatment groups received information by mail about wildfire safety and an invitation to schedule a free defensible space

1. EM-DAT, CRED / UCLouvain (2025); National statistical organizations and central banks, OECD national accounts, and World Bank staff estimates (2025) – with major processing by Our World in Data.

2. Work documenting these frictions includes Kousky, Luttmer, and Zeckhauser (2006), Shafran (2008), Gallagher (2014), Deryugina (2017), McCoy and Walsh (2018), Fu and Gregory (2019), Bakkensen and Barrage (2021), and Baylis and Boomhower (2023, 2026).

3. Wildfires accounted for 7% of global total natural catastrophe losses between 2015 and 2024, up from 1% between 2005 and 2014 (Swiss Re 2025).

4. For example, 62% of homes studied by Meldrum et al. (2015) have overgrown or unmaintained vegetation within 30 feet and 94% have other combustibles (firewood, propane tanks, etc.) within 30 feet.

assessment through a pre-existing fire district program. The subsidy groups received the same information plus a financial incentive of between \$100 and \$500 to be paid out upon successful completion of a set of defensible space requirements.

We quantify the effects of these treatments using data from program records, limited on-the-ground property assessments, and measures of pre-treatment vegetation developed from fine-scale satellite imagery. The remotely-sensed measures provide a low-cost, spatially consistent way to identify homes with defensible space concerns. An independently interesting contribution of our study is to report correlations between satellite measures of defensible space (which are increasingly used in research and practice) and ground-truth inspections performed by expert ESF staff.

The primary outcome measure in both experiments is whether a household contacted their fire district to request a defensible space assessment. Both districts view these inspection programs as part of a long-term strategy to engage residents by encouraging low-cost actions (e.g., removing leaves under wooden decks, relocating woodpiles away from exterior walls, cutting dry grass) in the near term while building interest in higher-cost actions (e.g., relocating large trees or shrubs near the home, removing tree limbs up to a specified height above the ground) going forward.

These free defensible space assessments were available to all residents prior to and throughout our experiments. However, contact rates in the control group are nearly zero. The information-only interventions see slightly increased response rates relative to the control groups. By contrast, we observe economically and statistically significantly higher response rates from those who received financial incentives to participate. A moral suasion intervention highlighting the need to reduce the risk to one’s neighbors was ineffective in our sample.

We also consider how the effect of the subsidy varies with household characteristics. In JCFD3, occupants of higher-value homes and homes in less socially-vulnerable areas are more likely to respond. In ESF these differences are not statistically significant. Baseline measures of wildfire risk including predicted annual fire probability and satellite measures of pre-treatment defensible space do not predict takeup in either experiment.

Finally, we use a small sample of on-the-ground re-inspections following the experiment to quantify improvements in defensible space. The confidence intervals around these estimates are large due to the limited number of inspections that were conducted during the study period and we cannot reject a wide range of possible effects.⁵

These results are relevant to the design and scale-up of adaptation incentive programs for wildfire and other disaster perils. Such programs are increasingly widely implemented

5. As we show, baseline to endline comparisons of the satellite measures are of limited usefulness for measuring treatment effects since the satellite measure is not sensitive to the low-cost actions incentivized in the experiment, which tend to be below canopy level and close to structure walls.

throughout the U.S. Examples (of many) include California’s Safe Homes grant program (Assembly Bill 888, 2025), income tax credits for wildfire mitigation in Colorado, and grants up to \$10,000 from the utility Pacific Gas & Electric.⁶ Beginning in 2024 (after our experiments were complete) the Oregon State Fire Marshal (OSFM) began offering a \$250 subsidy statewide to homeowners in high-risk areas who opt in to a defensible space inspection program.⁷

These results may also speak to the potential to drive mitigation investment through finer-grained homeowners insurance pricing. Technological progress and regulatory reforms are increasingly enabling property insurers to factor defensible space and other mitigation investments into premiums, though such risk-based pricing is still imperfect (Boomhower et al. 2024; Boomhower and Fowlie 2026). The extent to which such reforms would change investment in defensible space depends on the kinds of behavioral elasticities that we study in these experiments. We show that the subsidy amounts that we offered were roughly comparable to back-of-the-envelope estimates of annual premium discounts that might be offered for defensible space in a competitive insurance market. Subject to important caveats about contextual differences between insurance bills and public subsidy offers, this makes our estimates useful for thinking about how homeowners might respond to differentiated insurance pricing. This aspect of our results is made more relevant by the dearth of well-identified empirical estimates of these “adaptation elasticities.”

Our experiments build on existing research on self-protection from wildfire and other natural disasters. Much of this previous work is qualitative and survey-based, reporting correlations between survey responses, homeowner characteristics, and reported or observed mitigation behavior. These studies identify a range of factors including risk perception, neighbor spillovers, social context, aesthetic preferences, past wildfire experience, and financial barriers as possible factors affecting these self-protection decisions (e.g., Brenkert-Smith, Champ, and Flores 2006; McGee, McFarlane, and Varghese 2009; Brenkert-Smith, Champ, and Flores 2012; McCaffrey et al. 2011; Olsen et al. 2017). A small number of previous studies have used randomized experiments to measure homeowners’ visits to an informational website about wildfire risk (Meldrum et al. 2021; Byerly Flint, Champ, et al. 2022; Byerly Flint, Cada, et al. 2022). We are not aware of previous studies that have experimentally varied incentives for participation in defensible space programs or that have measured program participation or creation of defensible space.

We organize the paper as follows. Section 2 describes the empirical context, Section 3

6. California: https://leginfo.ca.gov/faces/billNavClient.xhtml?bill_id=202520260AB888; Colorado: <https://tax.colorado.gov/income-tax-topics-wildfire-mitigation-measures>. Incomplete lists of state and local programs are available at <https://www.resilienthomefund.org/programs/> and <https://www.smarthomeamerica.org/>.

7. See here for OSFM policy details: https://www.oregon.gov/osfm/wildfire/Documents/Dec2024_OSFM_Defensible_Space_Assessment_Incentive_Reimbursement_Pilot_Program_Guidelines.pdf.

reports on the design of the experiments, Section 4 describes the data, Section 5 discusses findings, Section 6 compares our offered subsidies to stylized insurance premium discounts, and Section 7 concludes.

2 Context

Between 1990 and 2020, the area defined as the “wildland-urban interface” (WUI) — the area where homes and fire-prone vegetation overlap — has expanded by 31% and the total number of houses within that area has increased by 46% (Radeloff et al. 2023). At the same time, wildfires have also become more difficult to control due to a combination of increased fuel loads due to decades of aggressive suppression policies (Littell et al. 2009; Hessburg et al. 2021) and climate change (Jolly et al. 2015; Abatzoglou and Williams 2016; Burke et al. 2021). As a result of these factors, wildfires are now destroying far more structures than ever before (Headwaters Economics 2025). Models of future wildfire risk project continued increases in the number of homes exposed to wildfires (Kearns et al. 2022; Riley et al. 2025).

However, not all homes that experience wildfires are destroyed. Historically, around 60% of the homes included in wildfire perimeters survive, though this figure varies substantially across incidents and types of homes within incidents (Baylis and Boomhower 2026). A given home’s survivability depends on structural characteristics and proximity to potential fuels (Cohen 2000; Quarles et al. 2013; Syphard, Brennan, and Keeley 2017). Structural characteristics include roof and siding materials, screening of vents, protection of the eaves, and combustible decks and fences. These features are most easily “hardened” during construction. Although retrofits are possible, the cost to fully raise an existing home to modern standards can be prohibitively high in all but the highest-risk areas (Quarles and Pohl 2018; Baylis and Boomhower 2026).

By contrast, managing fuels around existing homes can be far less expensive (Barrett and Quarles 2024). Broadly speaking, “fuel” refers to any combustible material located near enough to the home to ignite it in a wildfire. Firebrands from nearby shrubs, trees, or even other structures can “leap” across surprisingly large distances to ignite homes or nearby receptive fuels, and most structures in wildfires are ignited in this way (as opposed to direct flame impingement). Existing evidence suggests that reducing the amount of combustible vegetation and other fuels adjacent to the home increases structure survival during wildfires (Syphard, Brennan, and Keeley 2014, 2017; Meldrum et al. 2022). The first five feet beyond the exterior walls (the so-called “Zone 0” area) are particularly influential (Insurance Institute for Business and Home Safety 2025).

2.1 Defensible space programs

The vast majority of homes that will experience wildfires in coming decades have already been built. Since fully retrofitting these structures to modern wildfire building standards is prohibitively expensive, efforts to protect these homes increasingly focus on lower-cost defensible space investments. Various jurisdictions across the United States are promoting defensible space in different ways. De jure mandates for vegetation management are common in local and state law, but de facto enforcement of these policies is rare. California’s forthcoming Zone 0 mandate may become a notable exception, though enforcement mechanisms for that policy remain unclear (Newsom 2025). A parallel set of policy initiatives aim to create price incentives by enabling home insurers to differentiate premiums based on defensible space. These rules remove regulatory barriers to risk-based pricing or in some cases (including California and Colorado) affirmatively require differentiated pricing based on defensible space and structure hardening (Ludington, Liao, and Walls 2025).

Partly due to political opposition to mandates and risk-based pricing, many jurisdictions have instead relied on information provision and opt-in subsidy programs, as documented in note 6. These voluntary defensible space programs are designed to support homeowners in reducing or eliminating fuels near the home. In Oregon, Senate Bill 762 (SB762) in 2021 directed the Oregon State Fire Marshal (OSFM) to define requirements for high-hazard properties (Oregon Legislative Assembly 2025). However, statewide enforcement of these requirements has been paused in favor of voluntary approaches.⁸ To encourage adoption, OSFM allocated Community Risk Reduction funds to encourage property owners to comply with defensible space recommendations. The programs we report on in this paper reflect part of that effort.

2.2 Study locations

2.2.1 Jackson County Fire District 3

Jackson County in southern Oregon has a population of about 200,000, about half of whom live in the city of Medford. According to the First Street Foundation, 97% of the 90,000 homes in Jackson County face at least some risk of wildfire in the next 30 years, and overall the county is rated as facing a “severe risk.”⁹ Our partner Jackson County Fire District 3 (JCFD3) covers an area north and west of Medford. Figure 1(a) shows the tax lots included in the JCFD3 sample in blue, with the background coloring reflecting annual wildfire probability as predicted by the Wildfire Risk to Communities project (Scott

8. <https://apps.oregon.gov/oregon-newsroom/OR/OSFM/Posts/Post/oregon-adopts-defensible-space-model-code>.

9. https://firststreet.org/county/jackson-county-or/41029_fsid/fire.

et al. 2020). JCFD3 includes hilly, forested landscapes with high fuel density. This area has experienced numerous fires, including the 2020 Almeda and South Obenchain wildfires, which destroyed thousands of homes and burned tens of thousands of acres.

2.2.2 Eugene-Springfield Fire

After completing the JCFD3 experiment in 2023, we collaborated with Eugene-Springfield Fire (ESF) to run a similar experiment the following summer. ESF is located in Lane County in western Oregon. The city of Eugene occupies most of the jurisdiction. Figure 1(b) maps the ESF study area, which includes hilly, forested areas on the southern periphery of Eugene and Springfield (not pictured). ESF faces lower overall wildfire risk than JCFD3, but the area has experienced recent wildfires – most notably, the 2020 Holiday Farm Fire in Eastern Lane County that destroyed over 700 structures. The western edge of the fire was 14 miles from Eugene.

3 Experimental Design

This section describes the study samples, the treatment groups, and the timeline of each experiment.

3.1 Study Samples

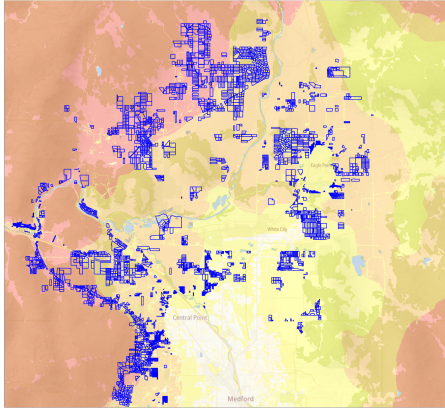
Jackson County. Working with JCFD3, we identified a sample of 4,662 single-family residential properties within the district boundaries that met a minimum level of wildfire risk.

Eugene. The ESF sample included 2,582 residential homes in relatively higher fire-risk areas of Eugene and Springfield. All were identified by ESF as facing some risk, and most were located in southern Eugene in or near forested areas.

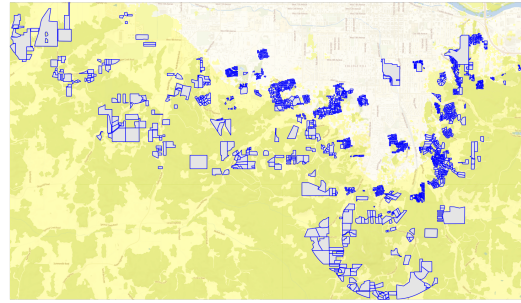
3.2 Treatment Groups

Households were assigned at random to either: (1) a control group that received no communication from the fire district; (2) an “Outreach” group that received information on wildfire risk and suggested actions to improve wildfire resilience, as well as information on the already-existing free home assessment program; or (3) one of two “Subsidy” groups that received the same information as the Outreach group as well as a financial incentive to be paid out upon successfully passing the home assessments. The offered subsidy amounts were \$250 and \$500 in JCFD3 and \$100 and \$200 in ESF. In JCFD3, all homes in the Subsidy treatment that passed the assessment were eligible for payment, even if no improvements to the property were required. In ESF, only homes where deficiencies were identified in the

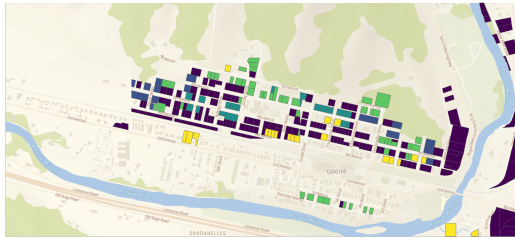
Figure 1: Study Areas



(a) JCFD3 Study Area

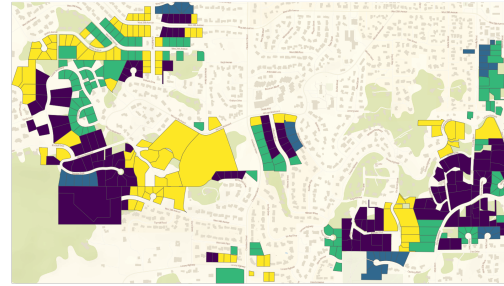


(b) ESF Study Area



Control Outreach Moral Suasion
\$250 Subsidy \$500 Subsidy

(c) Gold Hill (JCFD3)



Control Outreach
\$100 Subsidy \$200 Subsidy

(d) College Hill and Friendly Neighborhoods (ESF)

Notes: Maps of study areas. Top panels show the full extent of each study area with JCFD3 in Jackson County (left) and ESF in Lane County (right). Coloring in top panels indicates annual burn probability, where white is less than 0.001%, yellow is between 0.001% and 0.5%, orange is between 0.5% and 1%, and red is above 1%. Bottom panels show zoomed views of representative neighborhoods in each study area with parcels colored by treatment status.

initial inspection were eligible for a subsidy after correcting them. The differences in the amounts offered and payment rules reflected budget limits in each experiment and a desire to understand tradeoffs between offered subsidies and takeup.

The bottom panels in Figure 1 depict home-level treatment assignments in two representative neighborhoods. Gold Hill (panel c) is a densely populated area in JCFD3 along the Rogue River with high wildfire threat from adjacent forests, while the College Hill and Friendly neighborhoods (panel d) are typical of the ESF sample. Tax lots in each neighborhood map are colored according to assigned treatment group. To limit the number of near-neighbors receiving different treatment assignments, we randomized among home clusters, which we defined as sets of up to 10 adjacent homes on the same street.

JCFD3 included one additional treatment group, a “Moral Suasion” treatment that sent information but also emphasized the importance of defensible space to protect one’s neighbors and community.¹⁰ We did not include this treatment in the later ESF experiment due to its ineffectiveness relative to information alone. The number of homes assigned to each treatment arm in both experiments is in Table 1.

Table 1: Treatment Assignment

Treatment arm	Homes	Home clusters
<i>JCFD3</i>		
Control	3131	432
Outreach	216	27
Moral Suasion	181	27
\$250 Subsidy	579	81
\$500 Subsidy	555	81
<i>ESF</i>		
Control	984	156
Outreach	402	64
\$100 Subsidy	628	90
\$200 Subsidy	568	90

Notes: Number of homes and home clusters (the unit of randomization) by treatment arm for each experiment.

Finally, in addition to the primary treatment assignment, the JCFD3 experiment also incorporated a cross-cutting “Neighbor Nudge” treatment. Once a household in one of the treatment groups contacted JCFD3, their neighbors (other homes in the same home cluster) became candidates for an additional followup letter noting that their neighbors had been

10. This type of approach has been shown to be effective in other settings, e.g., in water conservation (Ferraro and Price 2013).

in touch with JCFD3 and encouraging them to get in touch as well. These follow-up letters were sent to all homes in a randomly selected 25% of relevant home clusters.

Households in the treatment groups were contacted via flyers or letters sent by postal mail. The mailings sent to each group were virtually identical aside from the changes needed to convey either the moral suasion element or the subsidy offer. The control group was not contacted. Copies of the contact materials are included in the appendix.

3.3 Timelines

The experiments were run during the summer and fall of 2023 (JCFD3) and 2024 (ESF). In general, initial communications were at the beginning of summer and households were counted as participating in the program if they got in touch with their respective fire district before the end of the summer.

For the JCFD3 experiment, we sent initial flyers in late April and early May 2023, with identical followup flyers two weeks later. In June, we expanded the number of households in the subsidy groups (drawing at random from the original control group) and sent all treated households the final followup letters in June and July. For the ESF experiment, we sent letters in batches starting at the end of July and followed up with a second mailing three weeks after each letter was initially sent. For ESF only, we arranged for ESF staff to perform a set of endline curbside assessments (discussed in detail below) between December 2024 and January 2025. Detailed timelines for both experiments are in the appendix.

4 Data

Sections 4.1 to 4.3 describe data on program participation, defensible space measurements, and auxiliary datasets. Section 4.4 describes the final analysis dataset and reports balance checks.

4.1 Program Participation and Assessments

In both experiments, the partnering fire district was responsible for tracking program participation as well as assessment requests and outcomes. For JCFD3, collection of program data followed the district’s existing process for defensible space assessments. Households requested assessments from the district via web form or phone. This contact was documented by the district and shared with the research team. JCFD3 staff then replied to the household to arrange an assessment within the next few weeks. Assessments were conducted following the Oregon Defensible Space Assessment checklist and logged using an online tool that recorded the home location and whether the home passed all eight items on the check-

list, as well as a free text field that included the recommendations given to the resident. The eight items covered the removal of flammable material under eaves, below decks, and within 100 feet of the home, guided the spacing of trees and plants, and required removal of firewood and barbecue tanks from near the structure. Appendix Figure A.4 includes a copy of the full checklist. Households in the subsidy groups were paid upon successfully passing an assessment, i.e., if the JCFD3 staff member did not identify any incomplete checklist items. For households ultimately receiving subsidies, the program data report whether the household passed on the initial assessment or a second followup assessment.

For ESF, the process was similar but facilitated by the use of Fire Aside, a commercial platform built to support defensible space clearance and maintenance among at-risk communities. Fire Aside managed most of the experimental operation, including sending information and subsidy offer letters, documenting responses from residents, and tracking assessment outcomes and subsidy payouts. At the end of the experiment, we received an export of the platform data from Fire Aside which included how and when each household was contacted, if and when they got back in touch with ESF, information about the assessment ESF staff conducted, and information on whether the household passed the assessment.

The ESF assessment was similar to JCFD3 but focused on a different set of requirements. The five recommended actions were: clear display of property address, clear roof and gutters, all vegetation cleared from first five feet around home, firewood at least 30 feet away from the home, and grass mowed to four inches or less. Homes in the subsidy groups were required to complete these five actions to receive the subsidy. Appendix Table B.1 lists additional inspection items that were recorded by inspectors but not required for subsidy payment.

4.2 Defensible space measurements

The program data include full in-person defensible space assessments for every person who requested an assessment (“participants”), but by construction do not include similar measures for non-participants or pre-treatment data for participants. We report on two methods for assessing defensible space in the broader study sample. The first is an on-the-ground “curbside” inspection wherein ESF staff visually assessed homes based on the view from the street. The second method relies on remote sensing. Both methods have limitations for measuring treatment effects: the curbside sample is small due to the high cost in terms of staff time, and the remotely sensed imagery is limited in its ability to measure *changes* in vegetation and other fuels targeted by our treatments (e.g., removing dead leaves or other flammable material from next to the house or trimming or removing ground-level juniper bushes that may be obscured by canopy vegetation or roof edges). We do find that these measurements are useful for understanding whether treatment effects on program participation vary according to pre-treatment vegetation density and for ground-truthing the fidelity

of remotely sensed vegetation measures, which is an independently interesting exercise given the growth of imagery-based approaches (Syphard, Brennan, and Keeley 2014; Mockrin et al. 2023; Collins et al. 2024; Escobedo et al. 2025; Tang and Fan 2025; Zamanialaei et al. 2025).

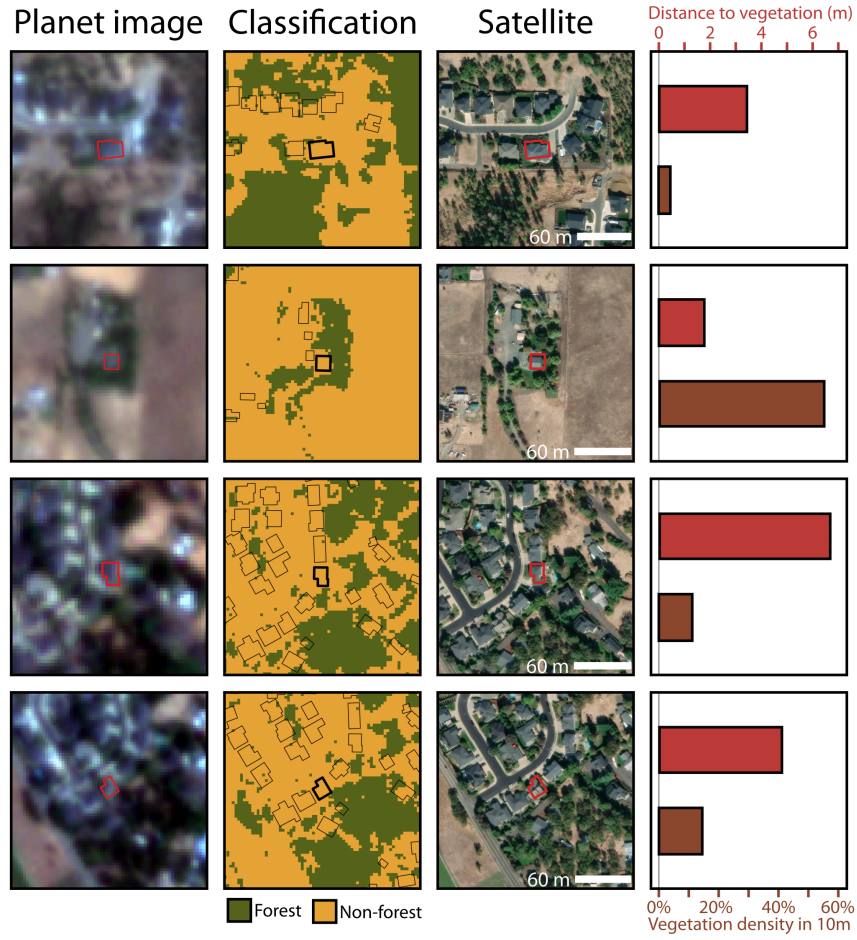
4.2.1 Curbside inspections (ESF)

ESF conducted 138 “curbside” assessments of homes chosen at random among the sample (without regard to treatment status), where an ESF staff member drove by the home and conducted a visual assessment of key defensible space characteristics without entering the property. These assessments included the same variables measured in the full inspections, but due to the limited access several items were recorded less frequently in these inspections (e.g., firewood presence, vegetation within 5 feet) since they could not always be observed from the front of the property. While conducting a full set of curbside assessments among the population of homes would have been prohibitively expensive in terms of staff time, this random sample allows us to test for impacts of the treatment beyond takeup itself. It also allows us to provide some “ground truthed” information to assess the ability of satellite imagery to measure vegetation clearance, as we detail in the next section.

4.2.2 Remote sensing measures

We complement the on-the-ground inspections with remotely sensed measures of defensible space. In many disaster mitigation settings, part of the challenge of encouraging defensive investments (e.g., storm shutters) is that verifying investments is costly, often requiring time-consuming in-person site visits and coordination with the resident. To demonstrate a potential solution to this problem in the wildfire context, we develop satellite imagery-based measures of defensible space. Section C provides full details on the way these measures were constructed. In short, we use multispectral imagery from the PlanetScope satellite constellation, which includes eight spectral bands at approximately 3 meters per pixel on a near-daily basis. We generate two cloud-free mosaics from images taken from dates before and after each experiment approximately one year apart in order to limit seasonal vegetation differences. We then run a supervised classification of the mosaics by training the model on locations with land cover classes that were stable across images. Finally, we use the classified mosaics alongside building footprint data for homes in the sample to produce the following remotely sensed measurements of defensible space: vegetation density in 3/10/30m, tree overhang (ESF only), and distance to vegetation. Figure 2 shows how two of these metrics (distance to vegetation and vegetation density in 10m) compare to high-resolution imagery for two homes in the sample.

Figure 2: Construction of Remote-Sensing Measurements



Notes: Figure showing original Planet imagery, classified imagery, comparison image from Google basemaps, and defensible space metrics (in columns) for example homes (in rows) in the JCFD3 area. The “Planet image” shows a true color composite of the Planet imagery for the end of the study period. “Classification” refers to the classified image, and “Satellite” shows a high resolution aerial image from Google (not necessarily taken during the same time period) for comparison. The final column shows two of the derived metrics for the given home.

The figure illustrates how the satellite imagery on the right compares to the machine classification of forest vs. non-forest pixels on the left, and the key defensible space metrics it produces are given in the bottom right.¹¹ It is worth highlighting that while the resolution of the satellite imagery is coarse, the classification leverages not only the visible spectral bands (used here to visualize the imagery), but all eight bands from the original PlanetScope imagery. As a result, the classification process is more accurate than might be supposed from the visible spectrum alone.

In Appendix Figure C.2, we measure the degree to which these satellite-derived metrics correlate with other observable data about the home (e.g., home value), the economic characteristics of the home’s Census block group, or model-based measures of landscape wildfire burn probability at that location. Burn probability has the largest correlation with the satellite metrics with values around 0.4 in the JCFD3 experiment, but all other correlations (including burn probability measured in ESF) are fairly weak. This suggests that the satellite measurement is capturing an important aspect of structure vulnerability that would not otherwise be detected from non-satellite data.

4.2.3 Comparison of on-the-ground and remotely sensed methods

As a secondary validation, we test the correspondence between the remotely sensed measures and the vegetation-related items in the 138 curbside inspections. The pairwise correlations are reported in Figure 3. Since the curbside inspections were conducted between December 2024 and January 2025, we use satellite metrics derived from imagery taken in October 2024.

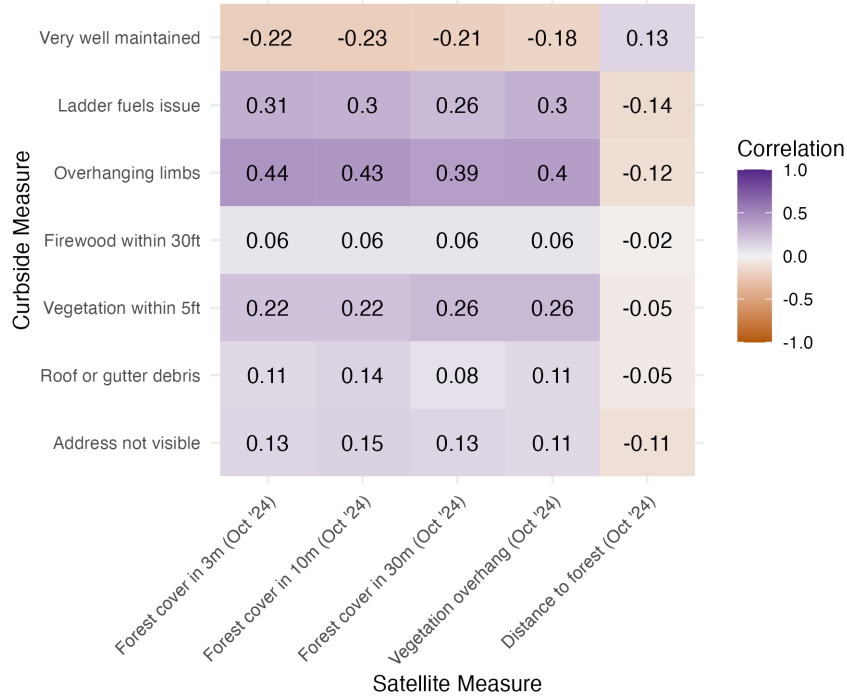
The figure shows that the curbside measures that correlate best with satellite imagery are those that represent issues likely to be viewable from above. Issues like having a non-visible address, roof or gutter debris, or firewood within 30 feet have correlations less than 0.15 in absolute value, while issues like ladder fuels and overhanging limbs have correlations between 0.26 and 0.44 in absolute value. Vegetation within 5 feet has a weaker correlation with satellite imagery, likely reflecting that the 3m pixel size of the underlying imagery is insufficient for detecting changes in fuels very close to the home. Still, keeping in mind the small number of homes with curbside inspections, we take this as supportive evidence that the satellite measures reflect on-the-ground vegetation.

4.3 Auxiliary data

County assessor data from Jackson and Lane counties were used to determine property class identifiers, home addresses (for mailing treatment letters), parcel acreage, and as-

11. Note that the imagery on the right is not the imagery used to train the model and should not be expected to match exactly to the metrics produced; it is higher-resolution imagery from Google Maps.

Figure 3: Correlations between defensible space measures



Notes: Figure visualizes correlations between satellite metrics of defensible space and curbside inspection issues related to vegetation for 138 homes in ESF that were selected at random for curbside inspections. Each cell is a pairwise correlation between the variables in that row and column. Each curbside measure is a row and each satellite measure is a column.

sessed home value. U.S. Census data at the Census block group (CBG) level were used to proxy for household characteristics, including proportion of owner-occupied homes, median age, proportion with a Bachelor’s degree or higher, and median earnings. We also use the Social Vulnerability Index (SVI), a CBG-level metric ranging from 0–100, where 100 indicates “more vulnerable”. The SVI is generated from 15 Census American Community Survey (2016–2020) indicators including income, age, language, and housing type (Coughlan, Ellison, and Cavanaugh 2019).

Finally, we use predicted annual burn probability from the Wildfire Risk to Communities project (Scott et al. 2020). This simulation-based measure gives the annual likelihood of a wildfire burning at the 30m square cell containing each home. It is produced from topographic and fuel cover data from 2013 and 2014. We use this metric to capture local differences in wildfire hazard. It does not account for any home-specific differences such as building material or defensible space.

4.4 Descriptive statistics and balance

Appendix Table D.1 reports summary statistics for the JCFD3 sample. The average home sits on 3.8 acres of land, is worth around \$490,000, and has a Social Vulnerability Index (SVI) of 44. The average annual burn probability in the sample is 0.6%. Census data available at the block group level show that 82% of residents own their homes, the median age of residents is 51, around 24% of the population has a bachelor’s degree or higher, and median annual earnings are \$42,000. Randomization successfully balanced observable characteristics across treatment groups. As reported in Appendix Section D.2, standardized mean differences (SMDs) between the control group and each treatment group are generally small across all measured covariates.

Appendix Table D.2 presents summary statistics for the ESF sample. This group faces lower wildfire risk than in JCFD3, with an average annual burn probability of 0.005% vs. 0.6%. The average home is worth \$375,000 and sits on 1.2 acres of land (smaller than JCFD3). The ESF sample areas have lower social vulnerability (SVI of 31 versus 44 in JCFD3) and higher educational attainment, with 60% of residents holding bachelor’s degrees. Treatment assignment in the ESF experiment similarly achieved good balance across observable characteristics.

5 Findings

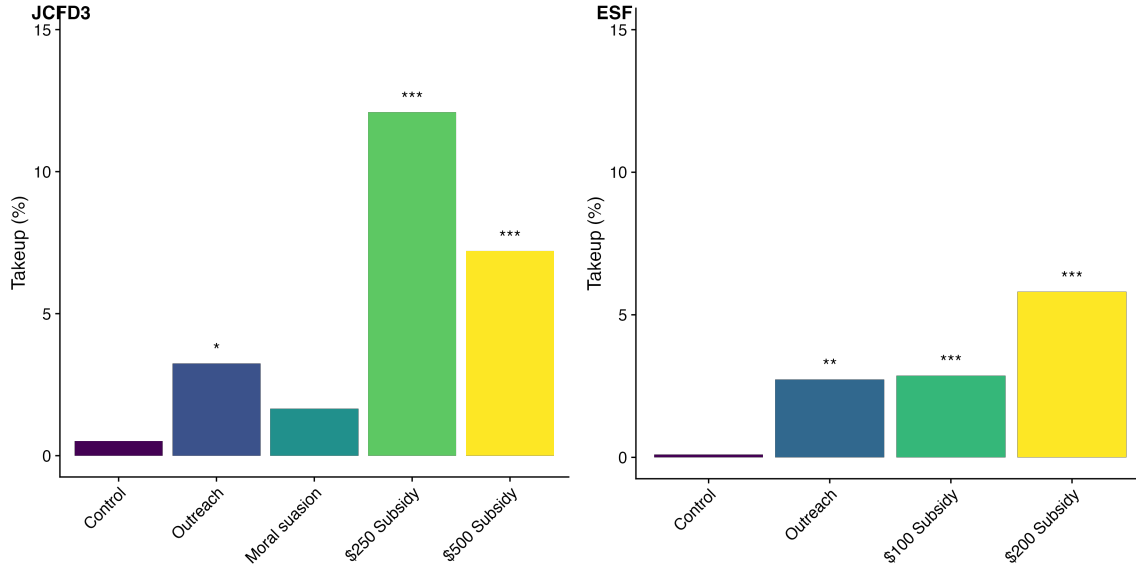
This section describes the findings of the experiments. Section 5.1 reports average treatment effects on assessment requests, Section 5.2 shows how the response evolved over time, Section 5.3 documents differential response rates by home or neighborhood characteristics, and Section 5.4 estimates the effects of the treatments on post-takeup outcomes.

5.1 Effects of Treatment on Contact

The primary outcome in this experiment is an indicator for whether households contacted their fire district during the treatment period to request a defensible space assessment. Figure 4 summarizes these takeup rates by treatment group. The bars show average takeup rates in each group, split by experiment.

The left panel shows estimates for JCFD3. Fewer than 0.6% of control homes, who did not receive a letter or a flyer, contacted JCFD3 for an assessment during the treatment period. This is consistent with the prior experience of JCFD3 staff. Homes that received the baseline information treatment contacted JCFD3 significantly more, although their contact rate was still low at about 3%. Less than 2% of the homes that received the additional moral suasion element in their contact materials requested assessments from

Figure 4: Takeup by Treatment Group

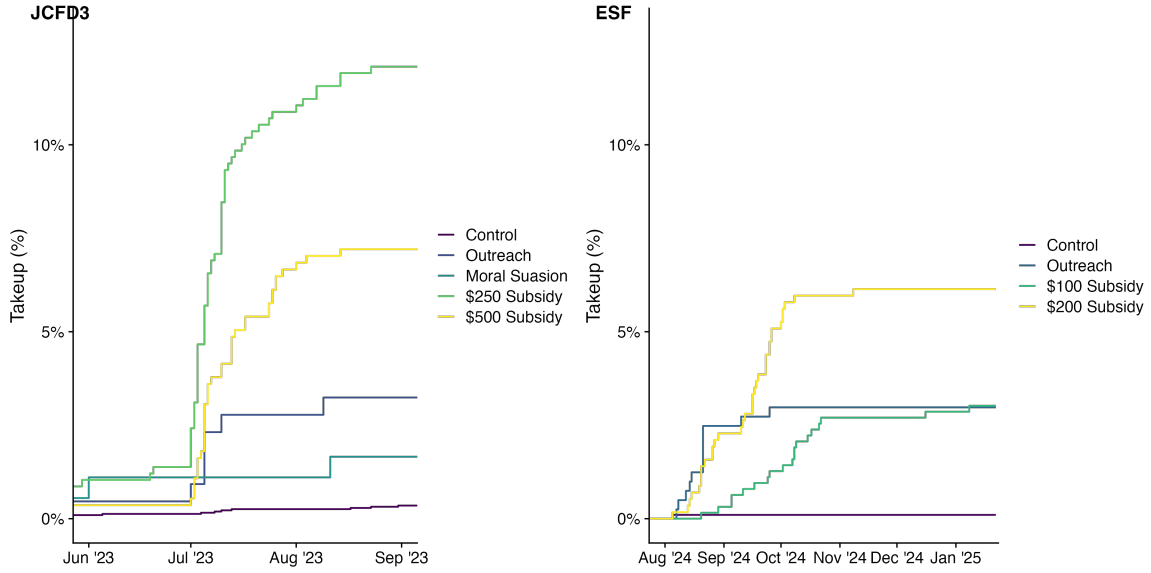


Notes: Figure shows average takeup rates by treatment group, split by experiment. The height of each bar is the mean takeup rate (in percentage points) among that group, where takeup is measured as contacting the respective fire district (JCFD3 or ESF) after the treatment period began. Left panel shows JCFD3 results, right panel shows ESF results. Stars indicate statistical significance for a t-test between the control group and the given treatment group. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

JCFD3, and their rate was not statistically distinguishable from the control group. Subsidy offers resulted in substantially higher contact. Twelve percent of the homes that received the \$250 subsidy offer requested assessments, a large and statistically significant increase relative to the control group. Finally, 7% of the homes in the \$500 subsidy offer group requested assessments. This estimate is also statistically different from the control group, and statistically different—and surprisingly, lower—than the contact rate among the \$250 group.

The right panel shows estimates for ESF. As with JCFD3, takeup among the control group is nearly zero. The outreach group sees takeup of around 3%, statistically different from the control group. Those receiving the \$100 subsidy offer also request assessments at a rate of around 3%, while those offered \$200 do so at around twice that rate, or 6%. Overall these response rates are remarkably similar to those we observe in JCFD3: both information groups responded at a rate of around 3%, and subsidy responses are similar considering the different amounts offered. Unlike JCFD3, we do not see the same counterintuitive relationship between the size of the subsidy offer and takeup—in ESF takeup is higher for the higher subsidy group.

Figure 5: Cumulative Takeup Rates



Notes: Figure shows takeup rates by date across both experiments, where takeup is measured as contacting the fire district after the treatment period began.

5.2 Response Timeline

To better understand how households responded to the treatment and to provide suggestive evidence on which types of contact seemed to be most effective, Figure 5 shows the cumulative takeup rates by group during the treatment period.

The left panel shows cumulative takeup rates for the homes in the JCFD3 sample. Response rates are relatively low early in the sample, when the team sent out the original flyers. In July, the response rates jump for all treated groups, and particularly for the subsidy groups. This is due to a combination of the addition of several hundred homes to the treatment group (discussed above) and the sending of the letter.¹² In other words, already-treated homes become more likely to respond upon receipt of the letter, and previously untreated homes also respond. These two factors contribute roughly equally to the increase in response rates. After August, few households got in touch with the fire district for the remainder of the sample period.

ESF shows a similar, though more gradual, pattern of takeup over time. Notably, the outreach-only group is actually most responsive during the first few weeks of the treatment, but the \$200 Subsidy group catches up by mid-September. It is only by the very end of the

12. One explanation for the larger effect of the letter on takeup rates than the flyers is that the letter was viewed as more legitimate by many households. Several individuals who requested an assessment commented that they found the letter, which was signed by JCFD3's Fire Chief, more credible than glossy flyers they originally received.

sample period (in January 2025) that the \$100 Subsidy group equaled the Outreach group. As in JCFD3, control responses remained low throughout the period, as expected.

5.3 Heterogeneity in Treatment Effects

This section considers differential effects of the intervention for different groups of households. We focus here on subsidies and combine the two amounts in each experiment into a single group (“Subsidy”) for parsimony. We estimate five separate versions of the following model:

$$\text{Takeup}_i = \alpha^S \text{Subsidy}_i + \delta H_i + \beta \text{Subsidy}_i \times H_i + \alpha^O \text{Outreach}_i + \alpha^M \text{Moral Suasion}_i + \varepsilon_i \quad (1)$$

where H_i stands in for above-median lot size, above median annual burn risk, above-median assessed value, above-median SVI, and above-median remotely-sensed vegetation cover within 10m, respectively.

Figure 6 visualizes the heterogeneity in treatment effects across both experiments, plotting α^S and $\alpha^S + \beta$ for each heterogeneity regression. In JCFD3, higher-value homes and homes in less socially vulnerable areas are substantially more responsive to treatment. The point estimates in the ESF experiment show the opposite with respect to home value and the same with respect to social vulnerability, though with somewhat noisier estimates due to the smaller sample size. Neither experiment shows statistically significant differences by lot size, wildfire hazard, or pre-fire vegetation proximity.

Finally, we examined whether the additional “neighbor nudge” treatment, where neighbors of individuals who contacted JCFD3 were sent an additional followup letter noting their neighbor’s participation, affected contact rates. Among the roughly 500 homes that received this additional nudge, we do not see any statistically significant changes in program participation. We report these findings in more detail in Appendix Section D.4.

5.4 Post-Assessment Outcomes

While the primary near-term goal of these programs is to initiate contact with residents and to assess their defensible space, the long-term objective is to increase resilience by reducing vegetation near the home. Assessment of vegetation clearance is challenging because it typically requires the physical presence of a member of the district’s staff—not to mention the agreement of the homeowner to permit access. A secondary goal of these studies was to pilot alternative approaches to assess vegetation clearance. To do so, we use the curbside assessment data collected for ESF and assessment pass rates from JCFD3.

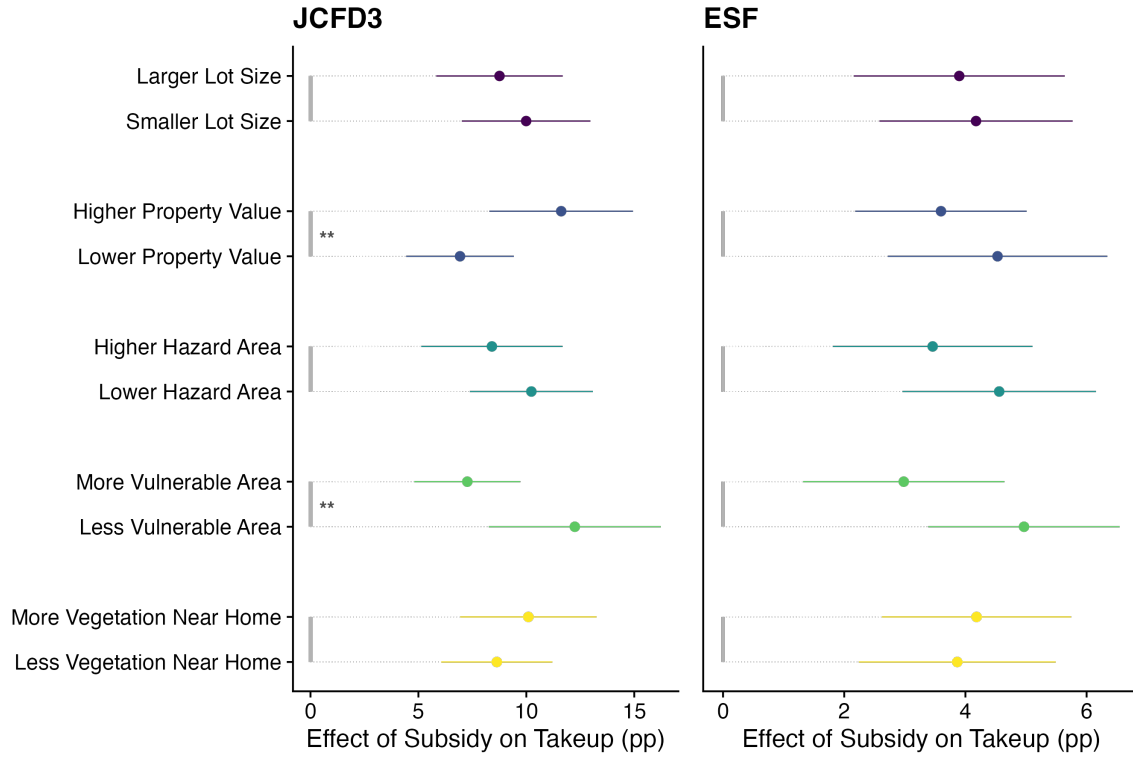


Figure 6: Heterogeneity in Subsidy Takeup

Notes: Figure shows effect of any subsidy offer on take-up probability (in percentage points) for households above and below the median values of lot size, home value, wildfire risk, social vulnerability, and proximity to vegetation (remotely sensed forest cover within 10 m). Models include controls for all heterogeneity dimensions plus city fixed effects. For JCFD3, models also include property class fixed effects. Standard errors clustered by street cluster. Error bars show 95% confidence intervals. Stars indicate statistically significant differences in subsidy effect between groups. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

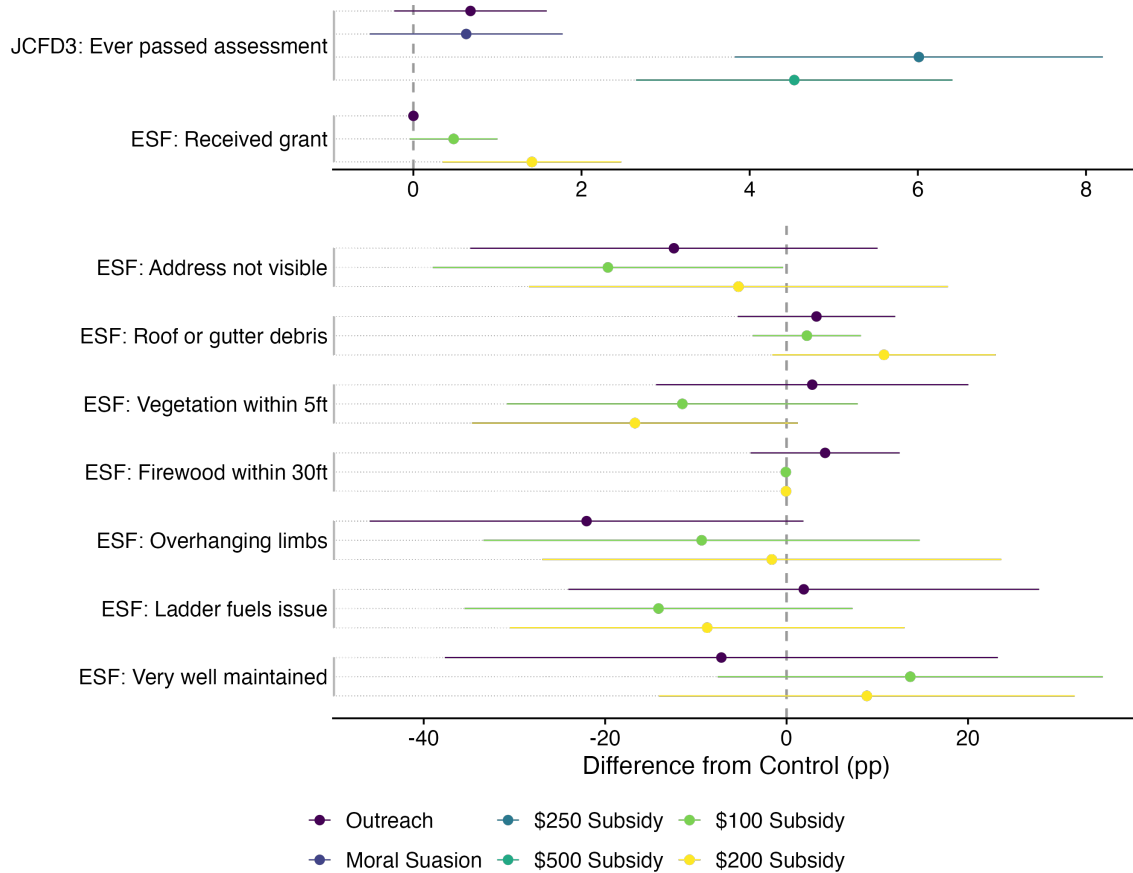
Figure 7 displays treatment effects on a range of post-assessment outcomes for both experiments. For JCFD3, we examine whether homes passed their defensible space inspection (either initially or after followup). For ESF, we look at grant receipt and curbside assessment outcomes (measured for a randomly selected subsample).

In JCFD3, we find that 6.0% (4.5%) of homes in the \$250 (\$500) subsidy group ultimately passed the inspection. Among those that passed, 60% (54%) did so on the first attempt. In total, slightly more than half of homes in the subsidy group that originally requested assessments in Figure 4 ended up passing: not all homes who failed the initial inspection ended up requesting a second. In the non-subsidy groups, around 0.5% of all homes in the group requested and passed the initial inspection, while the remaining 2.5% failed the initial inspection, received recommendations for improvement, and did not request a followup. In ESF, the proportion of subsidized homes that ultimately passed an inspection and received grants is much smaller; as described above, the ESF program only paid

out households who were a) in the subsidy group, b) identified as having issues to fix, and c) confirmed to have addressed those issues. This helped limit program costs by limiting grants to homes where payments were most likely to change defensible space outcomes.

The remaining estimates in the figure show intent-to-treat estimates of effects on the curbside assessment outcomes in ESF, which are only measured for 138 homes.¹³ The point estimates suggest that treated households may be less likely to have vegetation issues, but the confidence intervals for these outcomes are large and generally include zero.

Figure 7: Intent-to-Treat Post-Assessment Outcomes: JCFD3 and ESF



Notes: Figure shows treatment effects on intent-to-treat outcomes for both experiments. JCFD3 outcomes (top) include whether homes passed defensible space inspections. ESF outcomes (bottom) include grant receipt and curbside assessment results (measured on a random subsample of 138 homes). Points show coefficient estimates (in percentage points) from regressions of each outcome on treatment indicators with city fixed effects. Error bars show 95% confidence intervals with standard errors clustered by street cluster.

13. “Intent-to-treat” means that these estimates measure the effect of being assigned to a given treatment group, regardless of whether residents requested an assessment.

6 Subsidy Amounts Relative to Potential Insurance Discounts

In addition to opt-in subsidy programs, policymakers are actively debating the extent to which greater differentiation of home insurance pricing may be able to drive self-protection from wildfires. Technology companies and insurers have emphasized development of new rating methods and tools that will lower the costs of assessing and pricing defensible space structure-by-structure.¹⁴ Moreover, both California and Colorado have recently begun to require insurers to factor defensible space and home hardening into insurance premiums (Ludington, Liao, and Walls 2025). The effects of these price incentives on homeowner behavior depend on the kinds of behavioral elasticities measured in this study. Here, we compare the size of our subsidy offers to plausible estimates of insurance discounts for homes in our sample to understand what our experiments can teach us about potential homeowner responses to differentiated insurance pricing.

We adopt a stylized model of a competitive home insurance market where the wildfire component of the total annual insurance premium equals annual expected wildfire loss. In this model, the insurance premium difference δ_i for a given home i with vs. without defensible space is equal to the difference in expected loss:

$$\delta_i = L_i p_i \nu_i - L_i p_i (\nu_i - \tau) = p_i L_i \tau \quad (2)$$

where L_i is the loss cost (reconstruction cost), $p_i \in (0, 1)$ is the probability of wildfire exposure, $\nu_i \in (0, 1)$ is the probability of destruction conditional on a wildfire in the absence of defensible space (“vulnerability”), and τ is the reduction in vulnerability due to defensible space.¹⁵ We compute δ_i for every home in our sample, letting p_i be the annual probability of wildfire measured in Scott et al. (2020) and described above and L_i be each home’s assessed structure value. For τ , the reduction in vulnerability, we draw on Syphard, Brennan, and Keeley (2014), who measure that homes with 8–15 feet of defensible space were 12 percentage points more likely to survive a wildfire.

Table 2 reports these stylized premium differences for homes in the two study samples. On average, the implied annual price incentive to adopt defensible space in JCFD3 is \$200. One-fourth of homes have an annual benefit larger than \$256. Among participants in the

14. These include catastrophe models such as the Verisk Wildfire Model for the United States (Verisk Extreme Event Solutions 2025), categorical scoring tools like Zesty AI’s Z-FIRE <https://zesty.ai/products/wildfire>, and usage of aerial imagery for vegetation monitoring (Munce 2024).

15. This back-of-the-envelope calculation abstracts away from a variety of real-world details. We collapse a continuous range of possible fire intensities (e.g., flame lengths) and accompanying vulnerabilities into a scalar p_i and vulnerability ν_i . In practice, insurers pool homes into risk classes with prices reflecting average class-level loss risk as opposed to individual risk, so that discounts depend on class-average p as opposed to structure-specific p_i (Boomhower et al. 2024). We impose a constant effect of defensible space τ , ignoring potentially important interactions with home hardening.

program (households who contacted JCFD3 during our study period), those numbers are slightly higher. These single-year implied price incentives are comparable in size to the \$250 and \$500 subsidies offered in our experiment. While there are important contextual differences between state-sponsored opt-in subsidies and on-bill discounts from private insurers, we take the similarity in ballpark incentive amounts to mean that our experimental estimates are broadly useful for contemplating potential homeowner responses to differentiated insurance pricing.

Table 2: Stylized Insurance Discounts for Defensible Space

Sample	Mean	SD	p25	p75
<i>JCFD3</i>				
Full Sample	\$200	\$315	\$27	\$256
Participants	\$220	\$278	\$32	\$320
<i>ESF</i>				
Full Sample	\$1.23	\$2.13	\$0.23	\$1.49
Participants	\$0.97	\$1.58	\$0.23	\$1.08

Notes: Table provides stylized estimates of insurance discounts for defensible space mitigation for homes in JCFD3 and ESF. Insurance discounts are the annual implied insurance discount for mitigation δ_i (see text), computed separately for each household i . “Full Sample” rows summarize discounts across all households in each experiment (including both treated and control households) and “Participants” rows summarize discounts only among households who contacted their fire district during the study.

In ESF, the implied financial incentive to adopt defensible space under differentiated insurance pricing is substantially smaller than the subsidies that were offered in the experiment. Appendix Figure D.2 shows that this difference is primarily driven by the lower modeled probability of wildfire occurrence in the ESF study area. Efforts by Eugene-Springfield Fire to promote defensible space and wildfire safety may represent a response to the downside risk of low-probability urban conflagration combined with projected future increases in wildfire hazard throughout North America.

Another point of comparison for the subsidy amounts we offered is to actual discounts in markets where insurers differentially price wildfire vulnerability reductions. Ludington, Liao, and Walls (2025) report that California home insurers offered discounts between zero and \$600 per year for wildfire safety investments in 2025, with an average maximum available discount of \$200. These maximum discounts require both home hardening and defensible space investments. Discounts for defensible space alone average about \$50 per year, compared to our one-time subsidy amounts of \$100 to \$500.

7 Conclusion

We use two randomized experiments to understand how financial incentives and information provision affect homeowner participation in defensible space programs. Our main finding is that information paired with subsidies of \$100–\$500 drives substantially more adoption than information provision alone or moral suasion. This evidence is relevant to the design of a rapidly growing number of wildfire resilience programs, and more broadly to adaptation incentive and outreach programs for a variety of natural catastrophe perils.

Our findings also suggest that deliberate targeting of subsidies could provide significant benefits. Physical and social vulnerability to wildfire vary substantially across households within our study areas. Our treatments had equal or smaller effects for high-hazard and socially vulnerable households. Programs that vary the size of the incentive available to reflect disaster mitigation priorities (e.g., by offering larger incentives to higher-risk households or lower-income households) may achieve greater impacts in the highest-need populations. We demonstrate how remote sensing can be used to identify high-need areas.

Beyond subsidy programs, these findings are also relevant for understanding how homeowners might respond to pricing of wildfire resilience through differentiated home insurance premiums. Rapid technological change in insurance rating methods and regulatory reforms are expected to soon lead to widespread integration of defensible space and other structure-level characteristics into insurance pricing. Economists and policymakers champion the potential for these price incentives to drive resilience, but there is very little evidence on the relevant behavioral elasticities. We show that the subsidy amounts that we offered roughly correspond to the annual premium discounts that many households in our study might be expected to receive for defensible space in a competitive insurance market. Given appropriate caveats about contextual differences between insurance bills and government subsidies, we see our results as useful evidence that there is potential for pricing to drive resilience behavior – especially given the dearth of empirical evidence on this question.

In the course of this research, we developed remotely sensed measures of defensible space for all homes in the study samples. We pair these measures with expert in-person assessments of 138 properties to provide novel ground-truth evidence on the potential to monitor defensible space remotely. We find that remotely sensed measures correlate reasonably well with on-the-ground expert assessments. However, the sensitivity of remotely sensed measures to the kinds of low-cost *changes* in defensible space targeted in our experiments was limited.

Finally, experimental and anecdotal evidence from the study underscores the importance of a credible messenger even when offering financial incentives. In our first experiment, takeup rates increased substantially after we switched from sending glossy flyers to letters

signed by the local fire chief. Similarly, anecdotal communications with community members in Jackson County suggest that the counterintuitive higher takeup of the \$250 as opposed to the \$500 subsidy may have reflected a belief that the larger amount was a “too-good-to-be-true” offer.

Further research by social and natural scientists is important to refine the design and targeting of defensible space and home hardening subsidies. Stronger physical science and engineering evidence on the precise contribution of different elements of defensible space (e.g., canopy clearance vs. ground-level fuels) could allow for more specific program goals. Social science evidence on the longer-run effects of this type of program is also important for understanding the degree to which low-cost initial efforts of the kind encouraged by this program eventually lead to higher-impact home hardening investments and community-level mitigation projects. Relationship-building and long-run behavior change were central motivators for our fire district partners in offering defensible space programs.

More broadly, the effort to limit the growing threat from wildfires and other natural disasters will require policymakers to continue to evaluate how disaster resilience programs can better support residents of high-risk communities. Thoughtful program designs that reduce the physical and behavioral costs of risk mitigation will be essential to limiting the human and economic losses from such disasters.

References

- Abatzoglou, John T, and A Park Williams. 2016. “Impact of anthropogenic climate change on wildfire across western US forests.” *Proceedings of the National Academy of sciences* 113 (42): 11770–11775.
- Bakkensen, Laura, and Lint Barrage. 2021. “Going Under Water? Flood Risk Belief Heterogeneity And Coastal Home Price Dynamics.” *The Review Of Financial Studies* 35, no. 8 (November): 3666–3709.
- Barrett, Kimiko, and Stephen L. Quarles. 2024. *Retrofitting a Home for Wildfire Resistance: Costs and Considerations*. Report. Spring 2024. Bozeman, MT: Headwaters Economics, June.
- Baylis, Patrick, and Judson Boomhower. 2023. “The Economic Incidence of Wildfire Suppression in the United States.” *American Economic Journal: Applied Economics* 15, no. 1 (January): 442–73.
- . 2026. “Mandated vs. Voluntary Adaptation to Natural Disasters: The Case of U.S. Wildfires.” *Journal of Political Economy* Forthcoming.
- Boomhower, Judson, and Meredith Fowlie. 2026. “The Efficiency and Equity Impacts of Risk Classification in Catastrophe Insurance.” *Environmental and Energy Policy and the Economy* 7 (1): 3–32.

- Boomhower, Judson, Meredith Fowlie, Jacob Gellman, and Andrew Plantinga. 2024. *How are insurance markets adapting to climate change? risk selection and regulation in the market for homeowners insurance*. Technical report. National Bureau of Economic Research.
- Brenkert-Smith, Hannah, Patricia A. Champ, and Nicholas Flores. 2006. “Insights Into Wildfire Mitigation Decisions Among Wildland–Urban Interface Residents.” *Society & Natural Resources* 19 (8): 759–768.
- . 2012. “Trying not to get burned: understanding homeowners’ wildfire risk–mitigation behaviors.” *Environmental Management* 50 (6): 1139–1151.
- Burke, Marshall, Anne Driscoll, Sam Heft-Neal, Jiani Xue, Jennifer Burney, and Michael Wara. 2021. “The changing risk and burden of wildfire in the United States.” *Proceedings of the National Academy of Sciences* 118 (2): e2011048118.
- Byerly Flint, Hilary, Paul Cada, Patricia A. Champ, Jamie Gomez, Danny Margoles, James R. Meldrum, and Hannah Brenkert-Smith. 2022. “You vs. us: framing adaptation behavior in terms of private or social benefits.” *Climatic change* 174 (1): 11.
- Byerly Flint, Hilary, Patricia A. Champ, James R. Meldrum, and Hannah Brenkert-Smith. 2022. “Wildfire imagery reduces risk information-seeking among homeowners as property wildfire risk increases.” *Communications Earth & Environment* 3 (1): 229.
- Cohen, Jack D. 2000. “Preventing disaster: home ignitability in the wildland-urban interface.” *Journal of forestry* 98 (3): 15–21.
- Collins, L., L. Guindon, C. Lloyd, S.W. Taylor, and S. White. 2024. “Fractional cover mapping of wildland-urban interface fuels using Landsat, Sentinel 1 and PALSAR imagery.” *Remote Sensing of Environment* 308:114189.
- Coughlan, Michael R, Autumn Ellison, and Alexander Cavanaugh. 2019. “Social vulnerability and wildfire in the wildland-urban Interface.” *Northwest Fire Sci. Consort.*
- Deryugina, Tatyana. 2017. “The Fiscal Cost of Hurricanes: Disaster Aid versus Social Insurance.” *American Economic Journal: Economic Policy* 9, no. 3 (August): 168–98.
- Escobedo, Francisco J, Kamini Yadav, Onofrio Cappelluti, and Nels Johnson. 2025. “Exploring urban vegetation type and defensible space’s role in building loss during wildfire-driven events in California.” *Landscape and Urban Planning* 262:105421.
- Ferraro, Paul J, and Michael K Price. 2013. “Using nonpecuniary strategies to influence behavior: Evidence from a large-scale field experiment.” *Review of Economics and Statistics* 95 (1): 64–73.
- Fu, Chao, and Jesse Gregory. 2019. “Estimation of an equilibrium model with externalities: Post-disaster neighborhood rebuilding.” *Econometrica* 87 (2): 387–421.
- Gallagher, Justin. 2014. “Learning about an Infrequent Event: Evidence from Flood Insurance Take-Up in the United States.” *American Economic Journal: Applied Economics* 6, no. 3 (July): 206–33.

- Headwaters Economics. 2025. *Wildfires Destroy Thousands of Structures Each Year*. <https://headwaterseconomics.org/natural-hazards/structures-destroyed-by-wildfire/>. Accessed: 2025-12-08.
- Hessburg, Paul F, Susan J Prichard, R Keala Hagmann, Nicholas A Povak, and Frank K Lake. 2021. "Wildfire and climate change adaptation of western North American forests: a case for intentional management." *Ecological applications* 31 (8): e02432.
- Insurance Institute for Business and Home Safety. 2025. *Vegetation in Zone 0: Amplifying Damage to Structures*. IBHS Research Report. Insurance Institute for Business & Home Safety, December.
- Jolly, W Matt, Mark A Cochrane, Patrick H Freeborn, Zachary A Holden, Timothy J Brown, Grant J Williamson, and David MJS Bowman. 2015. "Climate-induced variations in global wildfire danger from 1979 to 2013." *Nature communications* 6 (1): 7537.
- Kearns, Edward J, David Saah, Carrie R Levine, Chris Lautenberger, Owen M Doherty, Jeremy R Porter, Michael Amodeo, Carl Rudeen, Kyle D Woodward, Gary W Johnson, et al. 2022. "The construction of probabilistic wildfire risk estimates for individual real estate parcels for the contiguous United States." *Fire* 5 (4): 117.
- Kousky, Carolyn, Erzo F. P. Luttmer, and Richard J. Zeckhauser. 2006. "Private Investment And Government Protection." *Journal Of Risk And Uncertainty* 33 (1): 73–100.
- Littell, Jeremy S, Donald McKenzie, David L Peterson, and Anthony L Westerling. 2009. "Climate and wildfire area burned in western US ecoprovinces, 1916–2003." *Ecological applications* 19 (4): 1003–1021.
- Ludington, Evan, Yanjun (Penny) Liao, and Margaret A. Walls. 2025. *From Risk to Reward: Insurance Discounts for Wildfire Mitigation*. Working Paper 25-30. Resources for the Future, December.
- McCaffrey, Sarah M, Melanie Stidham, Eric Toman, and Bruce Shindler. 2011. "Outreach programs, peer pressure, and common sense: what motivates homeowners to mitigate wildfire risk?" *Environmental Management* 48:475–488.
- McCoy, Shawn J., and Randall P. Walsh. 2018. "Wildfire Risk, Salience And Housing Demand." *Journal Of Environmental Economics And Management* 91:203–228.
- McGee, Tara K, Bonita L McFarlane, and Jeji Varghese. 2009. "An examination of the influence of hazard experience on wildfire risk perceptions and adoption of mitigation measures." *Society and Natural Resources* 22 (4): 308–323.
- Meldrum, James R., Christopher M. Barth, Julia B. Goolsby, Schelly K. Olson, Adam C. Gosey, James White, Hannah Brenkert-Smith, Patricia A. Champ, and Jamie Gomez. 2022. "Parcel-level risk affects wildfire outcomes: insights from pre-fire rapid assessment data for homes destroyed in 2020 East Troublesome Fire." *Fire* 5 (1): 24.

- Meldrum, James R., Hannah Brenkert-Smith, Patricia A. Champ, Jamie Gomez, Hilary Byerly, Lilia Falk, and Christopher M. Barth. 2021. “Would you like to know more? The effect of personalized wildfire risk information and social comparisons on information-seeking behavior in the wildland–urban interface.” *Natural hazards* 106 (3): 2139–2161.
- Meldrum, James R., Patricia A. Champ, Hannah Brenkert-Smith, Travis Warziniack, Christopher M. Barth, and Lilia C. Falk. 2015. “Understanding Gaps Between the Risk Perceptions of Wildland-Urban Interface (WUI) Residents and Wildfire Professionals.” *Risk Analysis* 35, no. 9 (September): 1746–1761.
- Mockrin, Miranda H., Dexter H. Locke, Alexandra D. Syphard, and Jarlath O’Neil-Dunne. 2023. “Using high-resolution land cover data to assess structure loss in the 2018 Woolsey Fire in Southern California.” *Journal of Environmental Management* 347:118960.
- Munce, Megan Fan. 2024. “Insurers are using satellite images to inspect homes in California — and dropping policies in the process.” *San Francisco Chronicle* (October).
- Newsom, Gavin. 2025. *Executive Order N-18-25*. Executive Department, State of California. Directs the Board of Forestry and Fire Protection to accelerate “Zone 0” ember-resistant zone rulemaking following the January 2025 Los Angeles and Ventura County fires. Sacramento, CA, February.
- Olsen, Christine S, Jeffrey D Kline, Alan A Ager, Keith A Olsen, and Karen C Short. 2017. “Examining the influence of biophysical conditions on wildland–urban interface homeowners’ wildfire risk mitigation activities in fire-prone landscapes.” *Ecology and Society* 22 (1).
- Oregon Legislative Assembly. 2025. *Senate Bill 762*. Oregon Legislative Information System. Regular Session.
- Quarles, Stephen L., Pam Leschak, Rich Cowger, Keith Worley, Remington Brown, and Candace Iskovitz. 2013. *Lessons Learned from Waldo Canyon: Fire Adapted Communities Mitigation Assessment Team Findings*. Technical report. Prepared for the Fire Adapted Communities Coalition. Tampa, FL: Insurance Institute for Business & Home Safety.
- Quarles, Stephen L., and Kelly Pohl. 2018. *Building a Wildfire-Resistant Home: Codes and Costs*. Research Paper. Prepared in partnership with the Insurance Institute for Business & Home Safety. Bozeman, MT: Headwaters Economics, November.
- R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing.
- Radeloff, Volker C, Miranda H Mockrin, David Helmers, Amanda Carlson, Todd J Hawbaker, Sebastian Martinuzzi, Franz Schug, Patricia M Alexandre, H Anu Kramer, and Anna M Pidgeon. 2023. “Rising wildfire risk to houses in the United States, especially in grasslands and shrublands.” *Science* 382 (6671): 702–707.

- Riley, Karin L, Scott N Zimmer, Evan Kodra, Isaac C Grenfell, Gregory K Dillon, Joe H Scott, Melissa R Jaffe, Julia H Olszewski, Kevin C Vogler, Mark A Finney, et al. 2025. “Spatial datasets of probabilistic wildfire risk components for the conterminous United States (270m) for circa 2011 climate and projected future climate circa 2047.”
- Scott, Joe H, Julie W Gilbertson-Day, Christopher Moran, Gregory K Dillon, Karen C Short, and Kevin C Vogler. 2020. “Wildfire Risk to Communities: Spatial datasets of landscape-wide wildfire risk components for the United States.”
- Shafran, Aric P. 2008. “Risk Externalities And The Problem Of Wildfire Risk.” *Journal Of Urban Economics* 64 (2): 488–495.
- Shields, Spencer G, Nicholas C Coops, Alexis Achim, Richard C Hamelin, and Christopher Mulverhill. 2025. “A review of PlanetScope CubeSats for forest monitoring.” *Science of Remote Sensing*, 100314.
- Swiss Re. 2025. *Wildfires*. <https://www.swissre.com/risk-knowledge/mitigating-climate-risk/wildfires.html>. Accessed: 2026-02-16.
- Syphard, Alexandra D, Teresa J Brennan, and Jon E Keeley. 2014. “The role of defensible space for residential structure protection during wildfires.” *International Journal of Wildland Fire* 23 (8): 1165–1175.
- . 2017. “The importance of building construction materials relative to other factors affecting structure survival during wildfire.” *International journal of disaster risk reduction* 21:140–147.
- Tang, Yongli, and Chao Fan. 2025. “Evaluating building-level tree cover change in Southern California wildland-urban interface using high-resolution satellite imagery.” *Journal of Environmental Management* 380:125160.
- Verisk Extreme Event Solutions. 2025. *Required Model Documentation for the PRID-2025-00001*. Model Documentation PRID-2025-00001. Verisk Analytics.
- Wright, Marvin N, and Andreas Ziegler. 2017. “ranger: A fast implementation of random forests for high dimensional data in C++ and R.” *Journal of statistical software* 77:1–17.
- Zamanialeci, Maryam, Daniel San Martin, Maria Theodori, Dwi Marhaendro Jati Purnomo, Ali Tohidi, Chris Lautenberger, Yiren Qin, Arnaud Trouvé, and Michael Gollner. 2025. “Fire risk to structures in California’s Wildland-Urban Interface.” *Nature Communications* 16 (1): 8041.

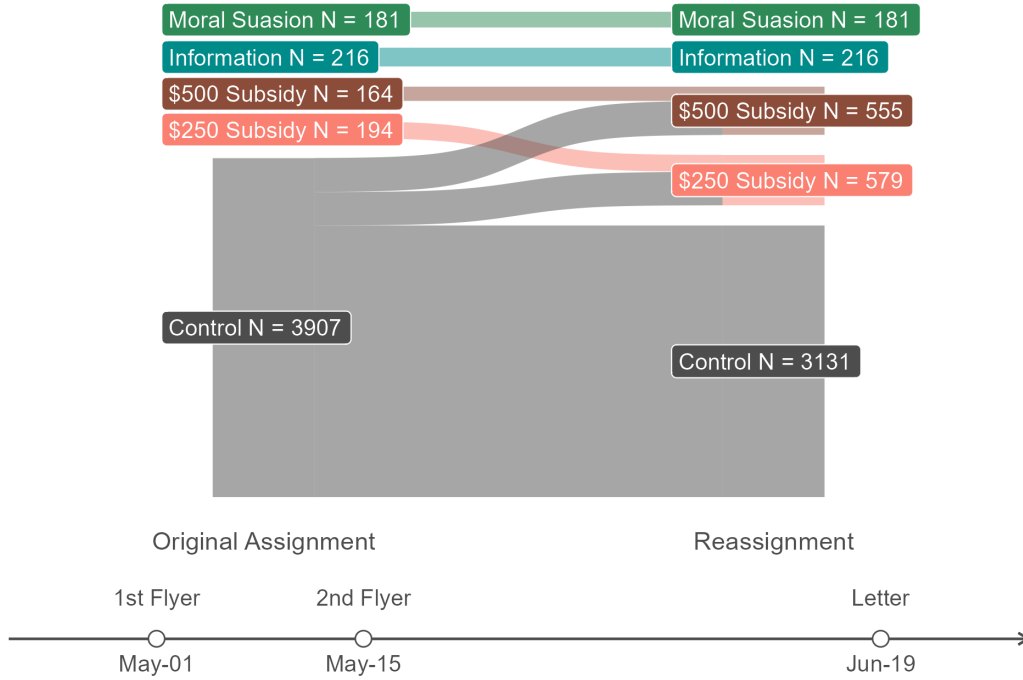
ONLINE APPENDIX

Online Appendix to “Incentivizing Self-Protection From Wildfires”

Patrick Baylis, Judson Boomhower, Bob Horton, and Chris Mulverhill

A	JCFD3 Experiment Implementation Details	A2
A.1	Timeline	A2
A.2	Contact Materials	A3
A.3	Assessment Materials	A5
B	ESF Experiment Implementation Details	A5
B.1	Timeline	A5
B.2	Contact Materials	A5
B.3	Assessment Materials	A8
C	Classification of Defensible Space With Satellite Imagery	A8
D	Supporting Exhibits	A11
D.1	Descriptive Statistics	A11
D.2	Balance Tables	A13
D.3	Reasons for Assessment Failure (JCFD3)	A14
D.4	Neighbor Nudge Effects	A15
D.5	Post-Takeup Outcomes (Including Satellite Measures)	A16
D.6	Heterogeneous Mitigation Premiums	A17

Figure A.1: Experiment Timeline



Notes: Figure shows timeline of experiment and randomization of homes into treatment groups for JCFD3. Initial assignment of homes is given by the boxes on the left and second assignment on the right. Originally treated homes remained in their treatment groups during the re-assignment, and a subset of control households were added to the \$500 and \$250 subsidy groups. Originally assigned treatment group homes received the first and second flyers as well as the letter, while the reassigned homes only received the letter.

A JCFD3 Experiment Implementation Details

This appendix section provides additional details about the implementation of the JCFD3 experiment. Section A.1 describes the timeline of the experiment, Section A.2 reproduces the printed contact materials, and Section A.3 gives details about the assessment procedure and requirements.

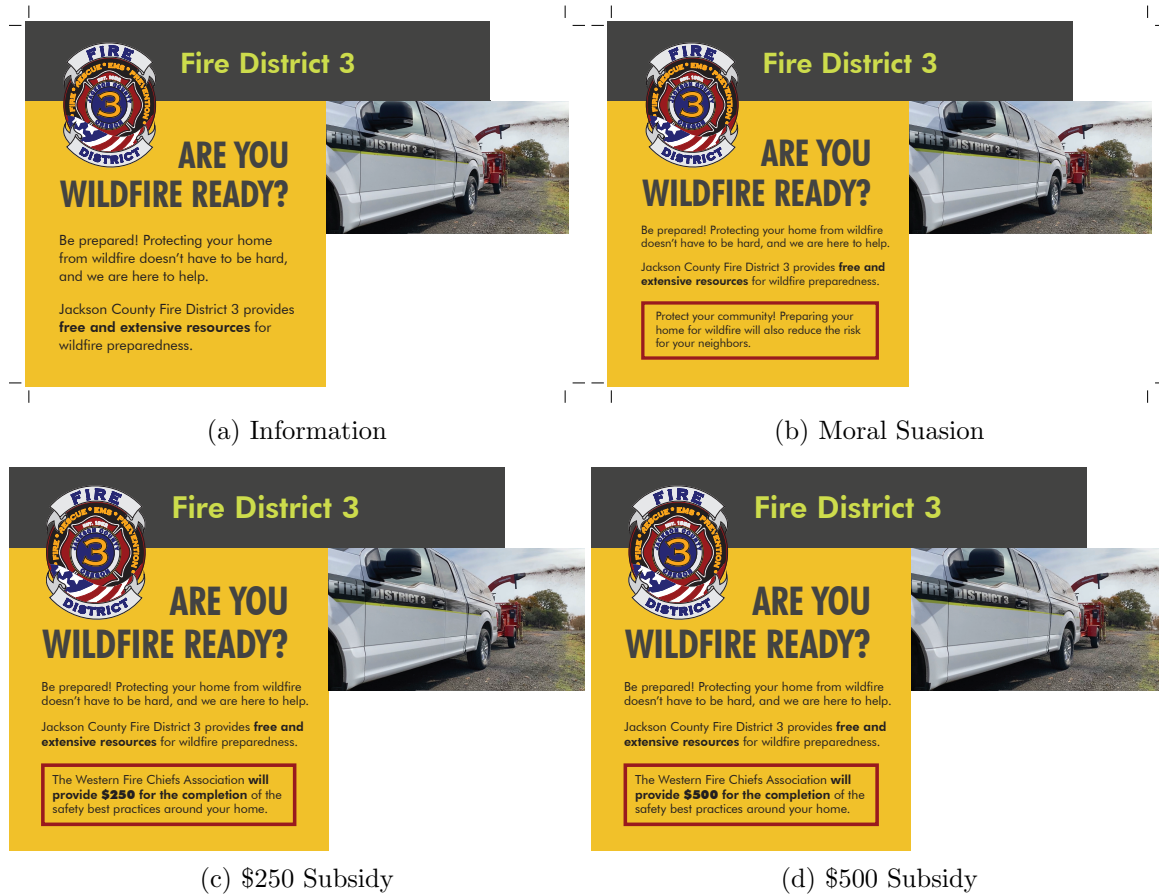
A.1 Timeline

The first round of contact was through flyers sent to each treated household in late April and early May, with identical followup flyers two weeks later, and a final followup letter in June and July. In June, since we had sufficient budget available, we increased the size of the treatment group by enrolling additional households at random from the control group into the subsidy groups. No additions were made to the Information or Moral Suasion groups. The analysis presented below considers a household to be in the treatment group if it was either assigned to that group initially or during the reassignment. A household is in the control group only if it was never assigned to a treatment group.

A.2 Contact Materials

This subsection reproduces the printed materials mailed to JCFD3 households. Figure A.2 shows the four treatment flyers, one for each treatment arm, which were sent twice, roughly two weeks apart. Figure A.3 shows the final followup letters sent in July; the Information and Moral Suasion groups received a common letter, while each subsidy group received a letter stating its subsidy amount.

Figure A.2: Treatment Flyers (JCFD3)



Notes: Figure shows flyers that were mailed to homes in treatment groups. Each treated household was sent the same flyer twice, roughly two weeks apart.

ONLINE APPENDIX

Figure A.3: Treatment Letters (JCFD3)

Are you wildfire ready? Jackson County Fire District 3 is here to help!

June 2023

Dear Jackson County Resident,

Wildfire season is just around the corner, and communities have to work together to reduce risk from wildfires. Jackson County Fire District 3 **offers free resources** to make it easier for residents to protect their homes.

These include the following:

- **Wildland home assessment** for recommendations about how to prepare your property
- **Fuels reduction trailer** delivered and picked up by FD3 to remove flammable vegetation
- **Community wood chipper** brought to your property and operated by trained FD3 crew to remove larger bushes and branches

Reach out today to request that Jackson County's risk reduction team assess your property, or to learn about other wildfire preparedness resources.

call (541) 826-7100 sign up by web form
email crn@jctd3.com jctd3.com/FuelsReduction



Safely maintaining vegetation and other materials around your home is one of the best things you can do to protect against wildfire, and communities are safest when everyone does their part. Reach out today to get wildfire ready!

Sincerely,

Mike Hussey
Fire Chief
Jackson County Fire District 3

(a) Information & Moral Suasion

Are you wildfire ready? Jackson County Fire District 3 is here to help!

June 2023

Dear Jackson County Resident,

Wildfire season is just around the corner, and communities need to work together to reduce risk from wildfires. Homes in your neighborhood are **eligible to receive \$250** for the completion of fire safety best practices.

Here's how it works:

- **Step One:** Schedule a **free assessment** of your property, and Jackson County Fire District 3's trained risk reduction team will let you know what you need to do to get wildfire ready. If your home is already wildfire ready, you will receive the \$250 benefit after the assessment.
- **Step Two:** Take the recommended actions to prepare your home for wildfire season.
- **Step Three:** The risk reduction team will conduct a **free follow-up assessment** and provide the \$250 benefit for completing recommended actions.

This one-time offer is funded by a grant program from the Office of the State Fire Marshal. The benefit payment comes from the Western Fire Chiefs Association and is available as long as funds last. Reach out today to request that Jackson County's risk reduction team assess your property, or to learn about other wildfire preparedness resources.

call (541) 826-7100 sign up by web form
email crn@jctd3.com jctd3.com/FuelsReduction



Safely maintaining vegetation and other materials around your home is one of the best things you can do to protect against wildfire, and communities are safest when everyone does their part. Reach out today to get wildfire ready!

Sincerely,

Mike Hussey
Fire Chief
Jackson County Fire District 3

(b) \$250 Subsidy

Are you wildfire ready? Jackson County Fire District 3 is here to help!

June 2023

Dear Jackson County Resident,

Wildfire season is just around the corner, and communities need to work together to reduce risk from wildfires. Homes in your neighborhood are **eligible to receive \$500** for the completion of fire safety best practices.

Here's how it works:

- **Step One:** Schedule a **free assessment** of your property, and Jackson County Fire District 3's trained risk reduction team will let you know what you need to do to get wildfire ready. If your home is already wildfire ready, you will receive the \$500 benefit after the assessment.
- **Step Two:** Take the recommended actions to prepare your home for wildfire season.
- **Step Three:** The risk reduction team will conduct a **free follow-up assessment** and provide the \$500 benefit for completing recommended actions.

This one-time offer is funded by a grant program from the Office of the State Fire Marshal. The benefit payment comes from the Western Fire Chiefs Association and is available as long as funds last. Reach out today to request that Jackson County's risk reduction team assess your property, or to learn about other wildfire preparedness resources.

call (541) 826-7100 sign up by web form
email crn@jctd3.com jctd3.com/FuelsReduction



Safely maintaining vegetation and other materials around your home is one of the best things you can do to protect against wildfire, and communities are safest when everyone does their part. Reach out today to get wildfire ready!

Sincerely,

Mike Hussey
Fire Chief
Jackson County Fire District 3

(c) \$500 Subsidy

Notes: Figure shows followup letters (the third mailing) that were mailed to homes in treatment groups. Note that the Information and Moral Suasion groups received the same followup letter.

A.3 Assessment Materials

JCFD3 inspectors used the standard Oregon State Fire Marshal defensible checklist to assess homes. That checklist is given in Figure A.4.

B ESF Experiment Implementation Details

This appendix section provides additional details about the implementation of the ESF experiment. Section B.1 describes the timeline of the experiment, Section B.2 reproduces the printed contact materials, and Section B.3 gives details about the assessment procedure and requirements.

B.1 Timeline

We conducted the original randomization of homes in June 2024. Each home in a treatment group received an initial mailing and a followup mailing. To limit the number of inspection requests coming in at the same time, we staggered the timing of mailings. 324 treated households were sent their first mailing on July 22, 2024. Another 578 households were sent their first mailings on August 6, 2024. The final round of first mailings were sent August 28, 2024. Followup mailings were sent approximately three weeks after the initial mailing.

Resident requested defensible space assessments were triggered by the resident getting in contact with ESF and scheduled following staff availability. The majority of these assessments took place between August and October 2024, with a handful occurring after. The randomized curbside assessments were conducted between December 2024 and January 2025.

B.2 Contact Materials

This subsection reproduces the printed materials mailed to ESF households. Figure B.1 shows the mailers for each of the three treatment arms. Treated households received the same mailer twice, with the followup sent roughly three weeks after the initial mailing.

Figure A.4: Oregon Defensible Space Assessment Checklist (JCFD3)

CHECKLIST

DEFENSIBLE SPACE

DATE _____

NAME _____

PHONE _____

ADDRESS _____

DEFINITIONS

DEFENSIBLE SPACE: Any area where combustible materials are treated, cleared, or modified to slow and reduce intensity of wildfire and allow space for fire suppression operations to occur.

FIRE-RESISTIVE PLANTS: Plants that can be used to reduce the likelihood of fire spread. See OSU's guide on page 1 for characteristics of fire-resistive plants.

LADDER FUEL: Low-hanging branches, leaves, needles, and other combustible plant matter that may allow wildfire to spread from low-growing plants to high-growing plants.

MEETS	RECOMMENDATION	DEFENSIBLE SPACE (100 ft out from the structure OR to property line, whichever is closer)
☐	☐	1) Flammable vegetation is removed from growing directly under the eaves. A minimum of five feet from the structure is recommended.
☐	☐	2) Leaves, conifer needles, deadwood, bark mulch, and other debris removed from the surface of, around, and below decks and fences.
☐	☐	3) Trees spaced and pruned following the example on page 2.
☐	☐	4) Leaves, conifer needles, deadwood, bark mulch, and other debris removed from within 100 feet of the structure or to the property line.
☐	☐	5) Fire-resistive plants are spaced within the designated defensible space area. Grass is mowed to less than four inches.
☐	☐	6) Firewood piles and lumber at least 30 feet from any structure.
☐	☐	7) Combustible vegetation 10 feet away from permanent propane tanks.
☐	☐	8) Small BBQ tanks, not in use, are stored at least 30 feet away or in an outbuilding.

RECOMMENDATIONS

PAGE 3

VERSION 1

Notes: Figure shows the defensible space checklist used to assess homes in the JCFD3 experiment.

ONLINE APPENDIX

Figure B.1: Treatment Letters (ESF)



Notes: Figure shows letters that were mailed to homes in treatment groups in ESF.

B.3 Assessment Materials

In ESF, inspections were conducted using the Fire Aside application. The application allowed the user to file a wide range of issues, but ESF inspectors primarily used a small subset of the possible issues. Table B.1 reports the inspection issues we focus on in this paper.

Table B.1: ESF Inspection Items

Measure	Description
Address not visible	Street numbers not legible from the road, which slows emergency response.
Roof or gutter debris	Accumulated dead leaf or needle litter on the roof surface or in gutters.
Vegetation within 5ft	Any live vegetation within five feet of the structure’s exterior walls.
Firewood within 30ft	A firewood pile stored within thirty feet of the home.
Grasses or clippings	Unmaintained dry grass and weeds, or piled grass clippings.
Overhanging limbs	Tree limbs extending over the roof or eaves.
Combustible fence	Combustible fence/gate attached to structure.
Ladder fuels	Ground-level vegetation beneath tree canopy that could carry fire from the ground into the crown.
Very well maintained	Inspector recorded the highest of three levels on the overall property impression scale.

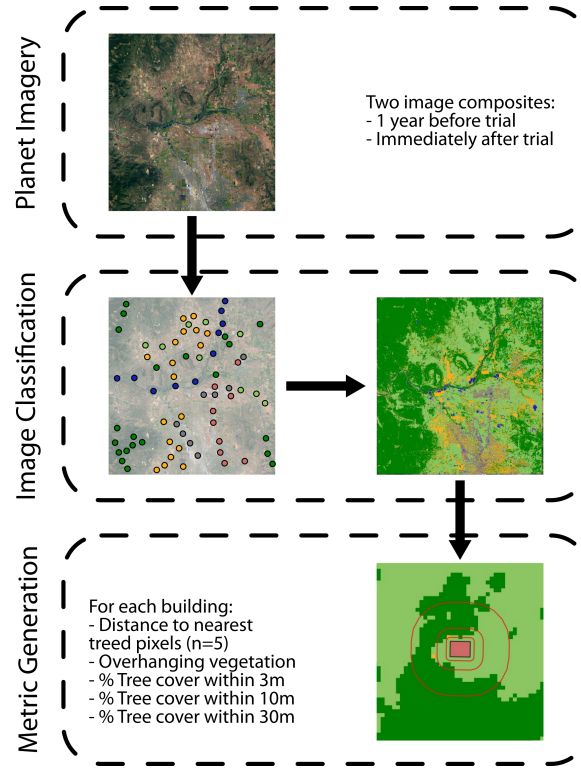
Notes: Table of inspection measures used for ESF. All measures are indicators, items in bold are those that were required in order to receive grant.

C Classification of Defensible Space With Satellite Imagery

This appendix section describes how we construct the satellite-based measures of defensible space used in the paper. We first describe the source imagery, then the supervised land cover classification used to distinguish treed from non-treed pixels, and finally the home-level vegetation metrics we derive from the classified imagery and building footprints.

We use multispectral satellite imagery from the PlanetScope constellation as the input data source due to its high spatial and temporal resolution: approximately 3 meters per pixel and near-daily revisit time (Shields et al. 2025). We downloaded cloud-free images, mosaicked them, and cropped them to each study area’s boundary. For each area, we used two image sets: one captured after the study’s completion and another from approximately one year earlier to reduce seasonal vegetation effects. For JCFD3, the images were taken on September 5, 2022 and October 5, 2023; for ESF, they were taken on October 6, 2023 and October 3, 2024. We acquired all images from the PlanetScope Dove constellation, which provides surface reflectance across eight spectral bands: Coastal Blue, Blue, Green I, Green, Yellow, Red, Red-edge, and Near-infrared.

Figure C.1: Workflow for remote detection of defensible space



Notes: Figure illustrates workflow for remote detection of building specific defensible space. Top panel illustrates source imagery used, middle panel shows the point classification process, and bottom panel describes the defensible space metrics generated using the classified satellite imagery and structure footprints.

We then used the mosaicked imagery to perform a supervised classification to delineate land cover types, focusing specifically on vegetation classes. The workflow for this process appears in Figure C.1. After mosaicking the images, we generated training samples by having a skilled interpreter manually identify and digitize polygons representing homogeneous areas for each of six land cover classes: Building/Bare, Forest, Shrub, Crop/Agriculture, Road, and Water. We visually inspected the training areas across multiple image dates to ensure that the selected pixels remained stable within their assigned class and did not change substantially between image sets, thereby minimizing misclassification due to temporal variability. In total, this process yielded more than 2.3 million sample pixels for JCFD and more than 1.3 million for ESF, distributed proportionally to the observed frequency of each class in the study areas.

We conduct the final classification using a Random Forest model.¹⁶ We train the model on spectral signatures extracted from the ENVI-generated training samples, using a random 70% of the samples for training and holding out the remaining 30% for accuracy assessment. The resulting classified raster provides a high-resolution land cover map for the full study area. After classifying each image, we condense the six land cover classes into two groups—

16. We use the Random Forest implementation in the ranger package in R (Wright and Ziegler 2017; R Core Team 2025).

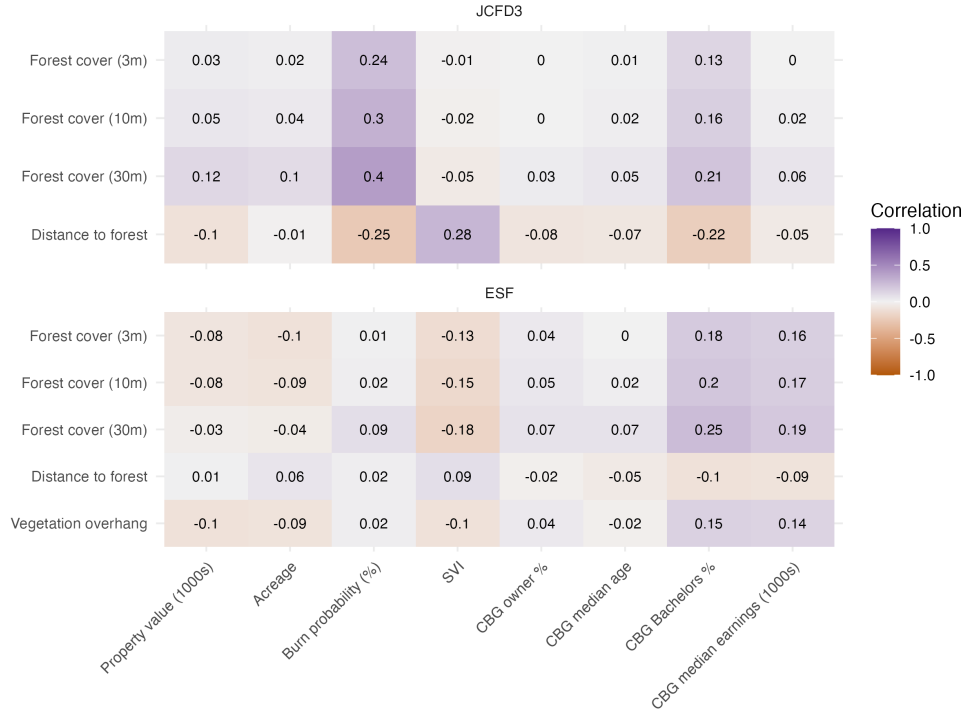
treed and non-treed—by merging all non-treed classes into a single category.

With the land cover map generated, we compute a series of descriptive vegetation metrics for the area surrounding each target home. We define the boundary of each home using the Microsoft building dataset and, when parcels contain multiple buildings, selected the largest building as the home. For each home, we calculated five metrics:

1. Zone 0 tree cover: Percentage of treed pixels within 3 m of building footprint (approximately zone 0)
2. Zone 1 tree cover: Percentage of treed pixels within 10 m of building footprint (approximately zone 1)
3. Zone 2 tree cover: Percentage of treed pixels within 30 m of building footprint (approximately zone 2)
4. Tree overhang: Percentage of the building footprint with treed overhang
5. Distance to vegetation: Average distance from the building boundary to the nearest five treed pixels

These metrics, following Syphard, Brennan, and Keeley (2014), were designed to capture both the density of nearby treed vegetation and the extent of defensible space around each property. The result was a quantitative, spatially and temporally explicit characterization of treed vegetation surrounding every home in the study. We focus on the distance to the nearest five treed pixels and the percentage of treed pixels within 30 m below, since the metrics correlate strongly with one another.

Figure C.2: Correlates of Remotely Sensed Defensible Space



Notes: Figure shows pairwise correlations between pre-treatment satellite-based vegetation measures (rows) and property/demographic covariates (columns), separately for each experiment (JCFD3 top, ESF bottom). Satellite measures use the earliest available pre-treatment image date for each experiment (September 2022 for JCFD3, October 2023 for ESF). Property value is assessed value in thousands of dollars; acreage is lot size; burn probability is the annual probability of wildfire at the home location; SVI is the Social Vulnerability Index for the home's Census block group (CBG); the remaining columns are CBG-level American Community Survey measures (owner-occupied share, median age, Bachelor's-degree-or-higher share, and median household earnings). JCFD3 does not have a vegetation overhang measure, so that row is blank in the top panel.

D Supporting Exhibits

This appendix section collects supporting tables and figures referenced in the main text. Section D.1 reports descriptive statistics for each study sample and Section D.2 reports covariate balance across treatment groups. Section D.3 summarizes which assessment criteria were most often unmet among JCFD3 homes that failed their initial assessment, and Section D.4 reports the effects of the neighbor nudge treatment. Section D.5 presents post-takeup treatment effects including the satellite-based vegetation measures, and Section D.6 shows the distributions underlying the stylized insurance premium calculations in Section 6.

D.1 Descriptive Statistics

This subsection reports summary statistics for the two study samples. Table D.1 covers JCFD3 and Table D.2 covers ESF, in each case summarizing takeover, property character-

ONLINE APPENDIX

istics, wildfire risk, Census block group demographics, and the satellite-based vegetation measures before and after the experiment.

Table D.1: JCFD3 Descriptive Statistics

Variable	Mean	P5	P95
Takeup	0.029	0	0
Ever passed assessment	0.014	0	0
Property value (1000s)	488	222	947
Acreage	3.8	0.14	10
Burn probability (%)	0.6	0.001	1.3
SVI	44	5.8	84
CBG owner %	82	72	93
CBG median age	51	37	67
CBG Bachelors %	24	6	41
CBG median earnings (1000s)	42	30	56
Forest cover in 10m (Sep '22)	23	0	79
Distance to forest (Sep '22)	16	2	63
Forest cover in 10m (Oct '23)	16	0	73
Distance to forest (Oct '23)	27	2.1	117
Change in forest cover	-5.4	-31	14
Change in distance to forest	11	-11	70

Notes: Table shows descriptive statistics for the JCFD3 sample. Burn probability is the annual probability of wildfire at the home location. Property value is assessed value in thousands of dollars. SVI is the Social Vulnerability Index for the Census block group (CBG) containing the home. CBG owner prop. is the proportion of people who own their homes in that CBG. CBG Median age is the median age of the Census respondent, CBG Bachelors+ prop. is the proportion of people with a Bachelor's degree or higher in the CBG, and CBG median earnings is the median household earnings in that CBG.

ONLINE APPENDIX

Table D.2: ESF Descriptive Statistics

Variable	Mean	SD	P5	P95
Takeup	0.024	0.15	0	0
Property value (1000s)	375	345	161	687
Acreage	1.2	7.8	0.12	5
Burn probability (%)	0.0049	0.0054	0.001	0.016
SVI	31	22	5.7	71
CBG owner %	79	11	60	91
CBG median age	50	5.5	43	58
CBG Bachelors %	60	17	28	78
CBG median earnings (1000s)	53	19	34	90
Forest cover in 10m (Oct '23)	81	26	25	100
Distance to forest (Oct '23)	2.6	1.9	1.7	3.7
Forest cover in 10m (Oct '24)	83	26	21	100
Distance to forest (Oct '24)	2.9	4.2	1.7	3.9
Change in forest cover	1.6	13	-22	25
Change in distance to forest	0.38	3.6	-0.54	0.67

Notes: Table shows descriptive statistics for the ESF sample. Takeup is an indicator for whether the household contacted ESF during the sample period. Property value is assessed value in thousands of dollars. Burn probability is the annual probability of wildfire at the home location. SVI is the Social Vulnerability Index for the Census block group (CBG) containing the home. CBG owner prop. is the proportion of people who own their homes in that CBG. CBG Median age is the median age of the Census respondent, CBG Bachelors+ prop. is the proportion of people with a Bachelor's degree or higher in the CBG, and CBG median earnings is the median household earnings in that CBG.

D.2 Balance Tables

This subsection assesses whether randomization balanced observable characteristics across treatment arms. Table D.3 and Table D.4 report control group means and standardized mean differences between each treatment group and the control group for JCFD3 and ESF, respectively.

ONLINE APPENDIX

Table D.3: JCFD3 Balance by Treatment Group

Variable	Control	Outreach	Moral Suasion	\$250 Subsidy	\$500 Subsidy
Property value (1000s)	491.8	0.19	-0.21	0.02	-0.14
Acreage	4.1	-0.08	-0.22	-0.03	-0.11
Burn probability (%)	0.6	-0.06	-0.16	-0.02	-0.11
SVI	42.5	-0.05	0.09	0.20	0.38
CBG owner %	82.4	-0.11	0.13	0.04	0.02
CBG median age	50.8	0.04	-0.22	0.01	-0.14
CBG Bachelors %	23.8	0.33	-0.23	-0.04	-0.13
CBG median earnings (1000s)	42.4	0.37	-0.10	-0.13	-0.18
Forest cover in 10m (Sep '22)	22.9	0.05	0.08	0.10	-0.12
Distance to forest (Sep '22)	16.3	-0.16	-0.15	-0.13	0.03

Notes: Table reports control group means and standardized mean differences (SMDs) for JCFD3 respondent covariates. SMDs are the differences in means between the given treatment group and the control group for that covariate, divided by the pooled standard deviation of the covariate in those groups.

Table D.4: ESF Balance by Treatment Group

Variable	Control	Outreach	\$100 Subsidy	\$200 Subsidy
Property value (1000s)	367.514	-0.03	0.01	0.09
Acreage	1.136	-0.05	-0.02	0.04
Burn probability (%)	0.006	-0.21	-0.26	-0.21
SVI	33.583	-0.23	-0.13	-0.22
CBG owner %	77.108	0.19	0.31	0.27
CBG median age	49.918	-0.09	0.05	0.12
CBG Bachelors %	57.664	0.18	0.26	0.31
CBG median earnings (1000s)	49.999	0.10	0.32	0.33
Forest cover in 10m (Oct '23)	81.231	-0.16	-0.06	0.15
Distance to forest (Oct '23)	2.553	0.03	0.06	-0.10

Notes: Table reports control group means and standardized mean differences (SMDs) for ESF respondent covariates. SMDs are the differences in means between the given treatment group and the control group for that covariate, divided by the pooled standard deviation of the covariate in those groups.

D.3 Reasons for Assessment Failure (JCFD3)

JCFD3 homes that did not pass their initial assessment received a list of recommended improvements, any one of which could have been the reason the home failed. Table D.5 reports, among these failing homes, the fraction that satisfied each individual criterion on the assessment checklist, indicating which requirements were most commonly binding.

Table D.5: Assessment Criteria Pass Rates Among Failing Households

Assessment Criteria	Fraction Passing
Flammable vegetation removal	0.63
Debris removal	0.42
Tree spacing	0.75
Debris spacing	0.67
Fire-resistant plants spacing	0.75
Firewood and lumber spacing	0.65
Combustible vegetation spacing	0.98
BBQ tanks spacing	0.96
Observations	57

Notes: Table reports proportion of households who passed each assessment criteria among households who did not pass the initial assessment.

D.4 Neighbor Nudge Effects

The final analysis we conduct focuses on the subset of homes who were eligible for the neighbor nudge treatment described in detail in Section 3.2. As a reminder, these are already-treated homes in the same home cluster as a neighbor who took up the treatment. Of these homes, around one quarter received a separate followup letter noting that one of their neighbors had contacted JCFD3 and inviting them to do so as well. Table D.6 documents the results.

Column (1) estimates the effect of a nudge without any additional controls. We do not see any additional effect of the neighbor nudge treatment on takeup. Column (2) includes controls for SVI, acreage, risk, and home value. Again, we do not observe a statistically significant effect of receiving a neighbor nudge.

Table D.6: Neighbor Nudge Treatment on Takeup

	Takeup (percentage)	
	(1)	(2)
Neighbor nudge	-0.310 (2.75)	1.53 (2.87)
SVI		-3.39 (5.21)
Acreage		0.027 (0.285)
Risk value		19.0* (11.1)
Assessment value		0.009 (0.006)
City FE		✓
Risk class FE		✓
Property class FE		✓
Observations	531	526
R ²	0.00003	0.04
Dependent variable mean	6.0	6.1

Notes: Table shows the effect of receiving the neighbor nudge treatment on takeup. The sample for these regressions is the set of homes who were eligible to receive an additional letter notifying them that their neighbors had been in contact with the fire district. The outcome variable is an indicator variable for the respondent contacting the district. Neighbor nudge is an indicator for whether they were actually contacted. SVI is the Oregon Social Vulnerability Index, Acreage is the lot acreage, Risk value is the wildfire risk, and assessment value is the assessed value of the parcel. Standard errors clustered by home cluster. *** p<0.01; ** p<0.05; * p<0.10.

D.5 Post-Takeup Outcomes (Including Satellite Measures)

This subsection extends the post-takeup results in Section 5.4 to the satellite-based outcomes. Figure D.1 repeats the intent-to-treat estimates shown in the main text and adds treatment effects on the remotely sensed vegetation measures for both experiments.

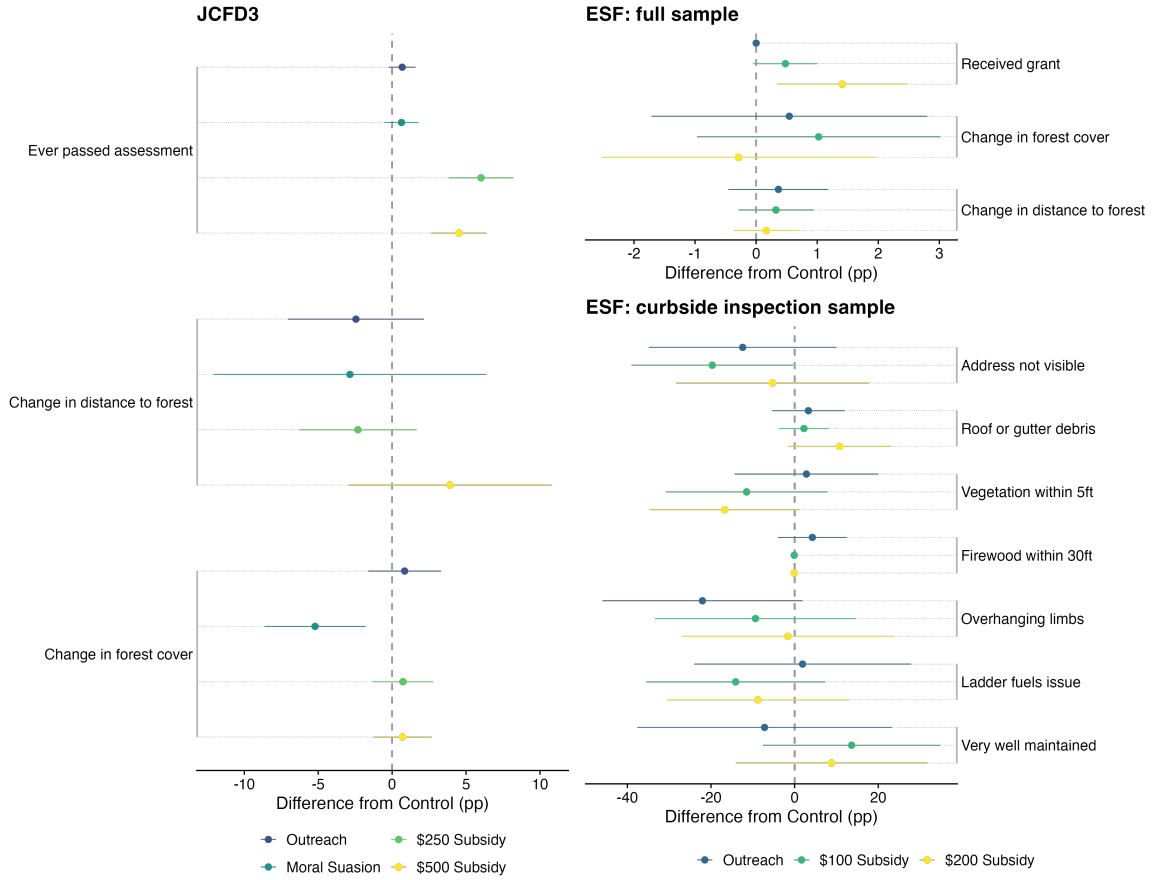


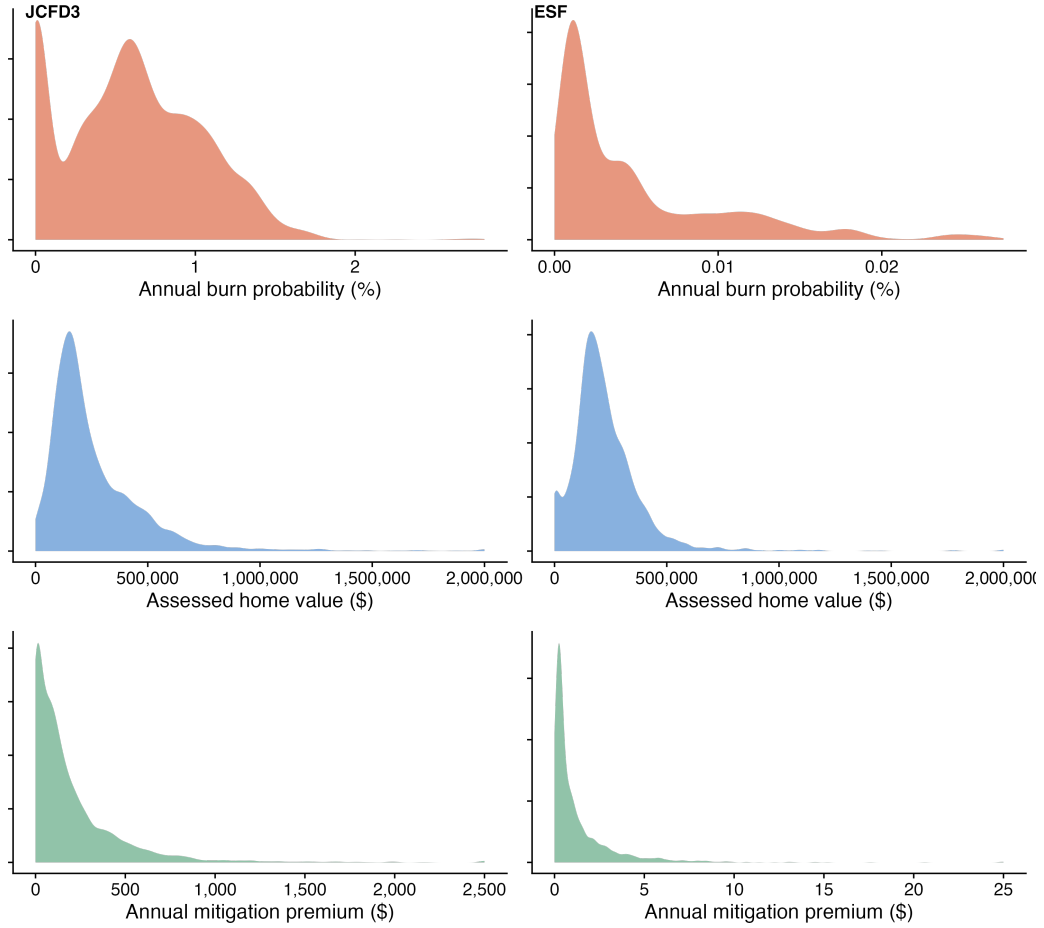
Figure D.1: Post-Assessment Outcomes: JCFD3 and ESF (All Measures)

Notes: Figure shows treatment effects on post-assessment outcomes for both experiments, including satellite-based vegetation measures. JCFD3 outcomes include inspection pass rates, distance to vegetation, and vegetation density in a 10m buffer. ESF outcomes include grant receipt, curbside assessment results, and satellite vegetation measures. Points show coefficient estimates from regressions of each outcome on treatment indicators with city fixed effects. Error bars show 95% confidence intervals with standard errors clustered by street cluster. Vegetation outcomes are from satellite imagery taken in October 2024.

D.6 Heterogeneous Mitigation Premiums

This subsection reports the distributions underlying the stylized insurance premium differences discussed in Section 6. Figure D.2 plots the annual burn probability p_i , the assessed home value L_i , and the resulting implied annual mitigation premium $p_i\tau L_i$ for each study area, showing that the smaller premiums in ESF are driven primarily by its lower modeled burn probability.

Figure D.2: Distributions of Burn Probability, Home Value, and Mitigation Premiums



Notes: Figure shows distributions of burn probability, home values, and stylized mitigation premiums in each study area. JCFD3 is the left column, ESF is the right column. Top row shows annual wildfire burn probability (p_i); middle row shows assessed value of the home (L_i); bottom row shows the expected annual mitigation premium ($p_i \tau L_i$, using $\tau = 0.12$ from Syphard, Brennan, and Keeley (2014)).

Appendix References

- R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing.
- Shields, Spencer G, Nicholas C Coops, Alexis Achim, Richard C Hamelin, and Christopher Mulverhill. 2025. “A review of PlanetScope CubeSats for forest monitoring.” *Science of Remote Sensing*, 100314.
- Syphard, Alexandra D, Teresa J Brennan, and Jon E Keeley. 2014. “The role of defensible space for residential structure protection during wildfires.” *International Journal of Wildland Fire* 23 (8): 1165–1175.
- Wright, Marvin N, and Andreas Ziegler. 2017. “ranger: A fast implementation of random forests for high dimensional data in C++ and R.” *Journal of statistical software* 77:1–17.